



Facial thermal imaging: A systematic review with guidelines and measurement uncertainty estimation

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ABSTRACT

The development of thermal imaging cameras has led to their widespread use, including for measuring facial skin temperature. However, the measurement protocol is not standardised, resulting in inaccurate and unreliable measurements. This systematic review of 315 papers selected according to the PRISMA guidelines aims to address this issue by reviewing how the papers reported key measurement parameters (type of thermal imaging camera, emissivity coefficient, measurement distance, ambient conditions, length of acclimatisation period) and how they analyse the thermograms. To make facial temperature measurements comparable within and between studies, we have proposed guidelines for accurate and reliable measurements and estimated measurement uncertainty for facial thermography as a result of this systematic review. Thus, a threshold of at least 0.3 °C is suggested to achieve a metrologically significant difference between experimental conditions or spatial or temporal points.

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1. Introduction

Non-contact infrared thermography has been used to measure skin temperature since the 1960s, with the main interest being the diagnosis of various medical conditions, such as breast cancer and malignant skin diseases [1]. The main advantage of contactless technology is that the measurement system is less intrusive, as the awareness of being measured can distort the interpretation of the measurement results [2,3]. Nowadays, thermal imaging is used to support diagnostics, monitor vital signs and study physiological pathologies, as well as to investigate emotions, assess anxiety and stress, evaluate mental workload, detect deception and other psychological measures [4–8].

To ensure the reliability of thermal imaging as a measurement technique, there is a tendency towards systematisation and standardisation, so that some organisations have issued their own guidelines for measuring skin temperature with a thermal imaging camera [9–16]. The biggest contributions to the regulation of the research field is the International Bureau of Weights and Measures (BIPM) “Best Practice Guide for Thermal Imagers” [17] and the consensus statement and checklist titled “Thermographic Imaging in Sports and Exercise Medicine (TISEM)” [18], both produced by leading experts in the field of thermal imaging of human skin temperature. Both documents focus on the key points of measurement, i.e. preparation of the room, the equipment and the person to be observed, as well as ensure that the operator records all relevant data [17,18]. One area of interest for human thermal imaging is temperature of the face, with some specificities (such as the topography of the face and the anatomy of the skin) that should be discussed separately. Thus, we reviewed the literature in the field of facial thermal imaging to highlight potential issues in measuring facial temperature with an infrared camera and to encourage researchers to standardise the research protocol according to the BIPM guide and TISEM.

In this review article, we first present the special features of facial thermometry from the perspective of the anatomy and topography of the facial skin (Section 2). We then introduce the operating principle of the thermal imaging camera (Section 3) and show how to perform an accurate measurement, from the selection of the thermal imaging camera and the right measurement protocol to the correct preparation of the person and the room (Section 4). We also present the analysis of facial thermograms and the challenges involved (Section 5). Furthermore, we suggest how to report appropriately on the measurements taken and the data analysed (Section 6). The review article concludes with a calculation of the measurement uncertainty of measuring facial skin temperature with a thermal imager (Section 7) and proposed guidelines for thermal imaging of facial skin (Section 8).

1.1. Literature search overview

In accordance with the PRISMA (Preferred Reporting Items for Systematic Reviews and Meta-analyses) guidelines for systematic reviews [19], we have included papers from the databases Scopus, Web of Science, PubMed, APA PsycInfo and Academic Search Complete (EBSCO). The focus of this review is on human facial thermography, so our search strategy used the following combinations of keywords: ((infrared camera) OR (infrared imaging) OR (infrared thermal imaging) OR (infrared thermography) OR (thermal camera) OR (thermal image) OR

Table 1
Overview of the papers included in the systematic review, organised by year of publication and the focus of their topic.

Year	Focus on physiology	Focus on psychology
2021–2023	[20–73]	[74–109]
2011–2020	[110–170], [171–193]	[194–275]
2001–2010	[276–300]	[301–309]
1991–2000	[310–318]	[319,320]
1980–1990	[321–332]	[333,334]

(thermal imager) OR (thermal imaging) OR (thermal imaging camera) OR (thermogram) OR (thermography) OR (thermovision)) AND ((face) OR (facial) OR (facial skin) OR (facial skin temperature) OR (facial temperature) OR (head) OR (head temperature) OR (psychophysiology) OR (vital)) AND ((adult) OR (children) OR (human) OR (people) OR (patient) OR (participant) OR (worker)) NOT ((NWIR) OR (near infrared) OR (SWIR) OR (animal) OR (infrared thermometer)). Inclusion criteria for eligible papers were that (i) the paper described an original experiment, (ii) the participants were human, (iii) the temperature of any part of the facial skin was measured, (iv) non-contact long or medium wavelength infrared thermography was used, and (v) the paper was written in English. Conference papers, review articles, letters, abstract-only publications and preprint papers were not considered. As we wanted to review all possible protocols and uses of the thermal imaging camera to measure facial temperature, we had no other specific exclusion criteria. We also reviewed the reference list of eligible papers and additional records that we identified through recommendations from academic platforms. These papers had to fulfil the same criteria to be included in the review. The review considered papers published online by the end of 2023. The entire process, from identification through to the final inclusion of studies, is detailed in the PRISMA flow diagram in Fig. 1, underscoring the systematic and transparent approach adopted in this review.

This literature search yielded 315 papers, which are listed in Table 1 and shown in Fig. 2 by year of publication and categorised according to whether the focus of the research is on pathology and physiology or on psychological variables. In the first category, we have also included studies that have investigated the technical aspect of using thermal imaging to measure facial skin temperature, as well as studies that have created databases of facial thermograms. Even if a paper described several different experiments, it was classified in one category according to the main objective of the paper. The ratio between the two categories is 2:3 in favour of papers focusing on physiology. Most (81.3 %) of the reviewed papers were published after 2010, which is consistent with the development of thermal imaging and the increasing use of this technique in various research fields. A detailed overview of the papers reviewed can be found in Appendix A.

2. Why is facial skin thermometry so ubiquitous?

It is clear from the reviewed papers that thermal imaging of the face is present in medicine as a diagnostic tool for the detection and research of pathology and physiological variables [22–56,110–158,276–291, 310–313,315–317,321–331], and monitoring of vital signals such as blood pressure [71,159,160], respiration (breathing) [57–65,161–174,

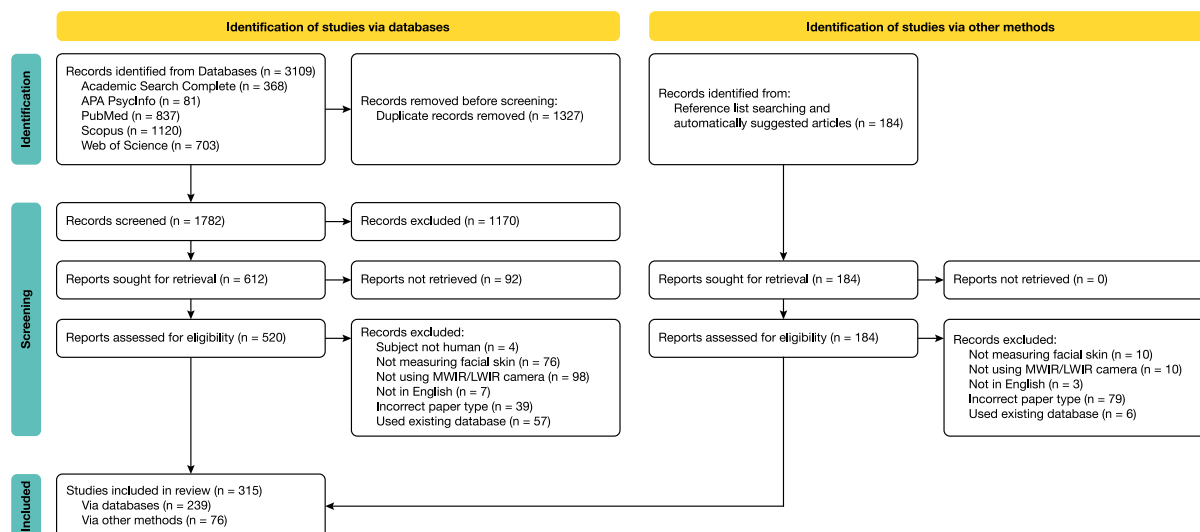


Fig. 1. PRISMA flow diagram, where n indicates number of papers relevant to each step.

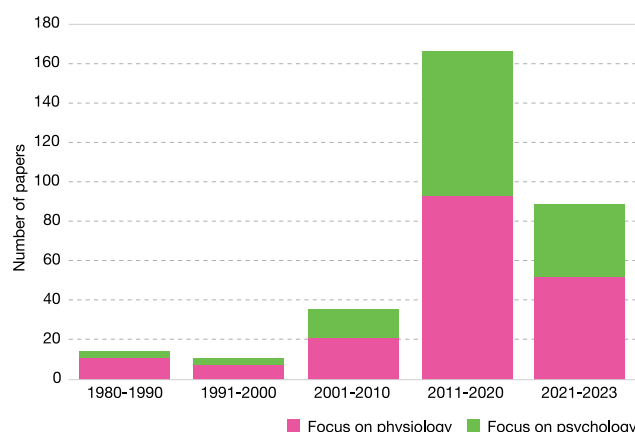


Fig. 2. Overview of the papers included in the systematic review, organised by year of publication.

297,298], and cardiac pulse (heart rate and variability) [67,68,161, 163,299]. Studies also explore facial surface temperature distribution and patterns in populations with and without medical conditions [69, 70,175–183,314,332], facial expressions [187,192,292,300] and in human impact studies (e.g. mobile phone use [188,293,318]). On the other hand, thermal imaging of the facial skin is also useful for psychological studies of mental, cognitive and emotional processes [74–109,194–275,301–309,319,320,333,334].

The facial skin not only protects the integrity of the body, prevents protein and fluid loss and repairs injuries, but also regulates body temperature [335]. The human body maintains a stable core temperature of about 37 °C using thermoregulatory mechanisms [336]. The central control of human thermoregulation is located in the preoptic/anterior hypothalamus in the brain, which triggers responses to dissipate heat (vasodilation and increased skin blood flow, sweating) when an elevated core or skin temperature is sensed, and to maintain heat (vasoconstriction and decreased skin blood flow) or generate heat (shivering) when temperature drops [337]. These processes of thermoregulation can be triggered by a change in core and skin temperature as well as by environmental conditions such as air temperature [337, 338]. The surface temperature of the skin can therefore fluctuate widely and is dependent on external factors such as the ambient temperature, relative humidity and convective air movement, the heat from the

body core delivered to the skin via the peripheral blood flow and the heat generated by the metabolic activity of the skin [336,338–340]. Therefore, the surface temperature of the skin is affected by various dermatological and medical conditions. The human body has a circadian rhythm and, in women, an additional rhythm determined by the menstrual cycle, both of which influence skin temperature [341–343]. The circadian rhythm of the proximal skin temperature, e.g. on the forehead, is highest in the evening (18:00) and lowest in the morning (10:00) [342] and follows that of the core temperature [343]. The core temperature is also 0.3–0.7 °C higher in the post-ovulatory luteal phase than in the pre-ovulatory follicular phase [343]. However, the measured skin temperature is typically lower than the core temperature (provided that the skin is not directly exposed to local heat sources), as the skin temperature can only approach the core temperature when the blood vessels are maximally dilated [343]. Facial skin blood flow regulation and sweating may also be influenced by cognitive and psychological processes, such as the fight-or-flight response, social interactions or emotional state [4,344–347]. Therefore, the measurement of facial skin temperature is also useful outside of medical pathology.

2.1. Facial skin anatomy

The first specific feature of facial skin that potentially affects the temperature measurement of a thermal imager is the unique anatomy of the skin and thus the surface of the face that is observed by the thermal imager. In general, the skin consists of epidermis, dermis and subcutaneous tissue [348] (Fig. 3). Since water absorbs infrared radiation, the radiation detected by the camera is emitted from the outermost protective layer of the epidermis, the *stratum corneum* (SC), and not from the underlying living tissues that are in a water environment [338]. The SC on the face is an extremely thin and soft biological membrane composed mainly of dead keratinocytes, known as corneocytes, which are much smaller and have a higher turnover rate than those on the extremities or trunk [349–351]. A high turnover rate and adequate hydration support the compact structure of SC and maintain the smoothness and softness of the skin surface [349]. The main role in the permeability of the SC and the integrity of the skin barrier function is played by the hydrophobic lipids in its intercellular spaces [352]. A decrease in the biosynthesis of SC lipids contributes to the alteration of the skin barrier function and the increased skin dryness that occurs during the winter months and with advanced age [340,352]. The effectiveness of the SC barrier function is assessed by measuring transepidermal water loss (TEWL) [349,352,353]. TEWL

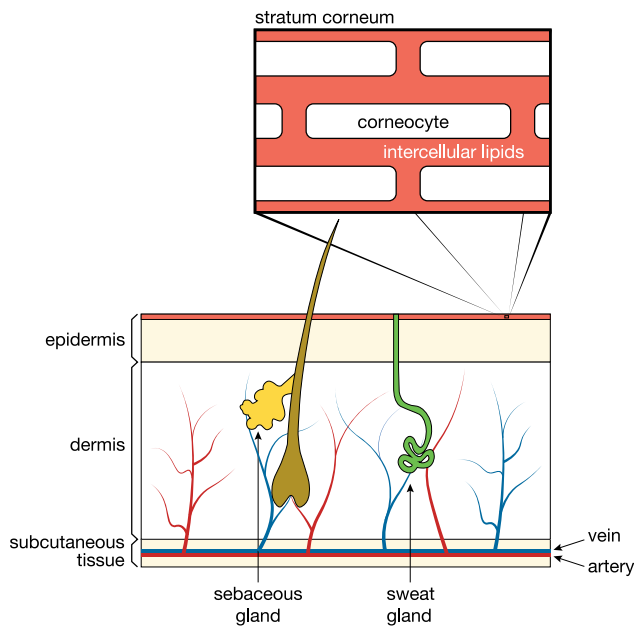


Fig. 3. A simplified illustration of the anatomy of the skin layers.

indicates the amount of water, without sweat, passing out of the epidermis, most of which is passive evaporation of water from the skin surface [352,353]. SC hydration does not have a time-dependent rhythm whereas TEWL does [342]. The highest TEWL values on the forehead are in the evening, around 20:00, the lowest in the morning between 8:00 and 10:00 [342]. The TEWL is also influenced by the hydration state of the SC, the surface temperature of the skin, the ambient temperature and the relative humidity [340,352,354]. In the northern hemisphere, both skin temperature and TEWL are highest in summer (July) and lowest in winter (January) [340]. Lower ambient temperatures and higher relative air humidity decrease TEWL, whereby the effect of humidity on the TEWL is more pronounced at higher temperatures [354]. Higher relative air humidity also has a positive effect on SC hydration [354,355], so that fine lines and a rougher skin surface are more accentuated in an environment with low air humidity [356]. The stability of the relative air humidity in the room is also important, as a 30 % difference in relative air humidity can lead to visible changes in the properties of the skin in 30 min [356]. Hydrated SC protects more effectively against UV rays [357], with frequent sun exposure reducing SC moisture [358].

The face is also rich in pilosebaceous glands, units of sebaceous gland and hair follicle, and eccrine sweat glands [349,350,359,360]. Both types of glands open on the surface of the skin and these openings form anatomical structures colloquially known as facial pores [361, 362]. Facial pores are flexible slight depressions on the surface of the skin which vary in size depending on genetic predisposition, ethnicity, age, gender, hormones, excess sebum, acne, skin elasticity, skin care and other factors [361–363]. The sebaceous glands produce sebum, a mixture of sebaceous lipids and cellular debris, which serves as a heat insulation layer and also as hydrophobic protection against overmoisture [364]. The level of sebum production is influenced by androgen hormones, age, diet, skin temperature, season, ambient temperature and relative air humidity [340,364–367]. The main function of the eccrine sweat glands is to produce sweat, a clear, colourless and odourless fluid enriched with organic and inorganic electrolytes, which evaporates and dissipates heat to regulate body temperature [365, 368,369]. Sweating transfers heat from the body to water on the surface of the skin and increases SC hydration [365]. The rate of sweating depends on the density of active sweat glands and the rate of secretion per gland, so sweating on the forehead, an area with

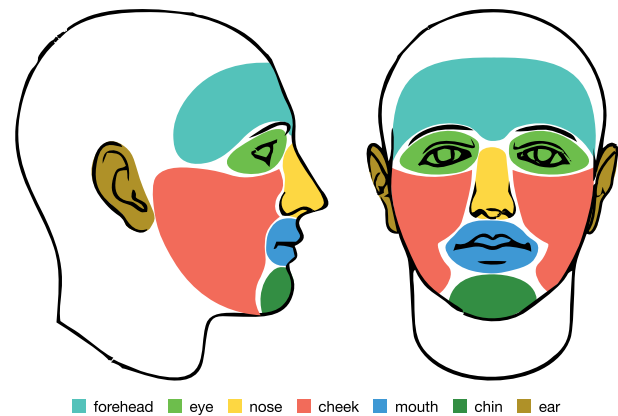


Fig. 4. The facial regions covered in this systematic review.

a high density of eccrine glands, can be very noticeable [345,365]. While the concentration of sweat glands is high on the forehead, there are none on the lips [345,369]. Sweat gland activity is influenced by a number of physiological and individual factors, such as circadian rhythm, menstrual cycle, skin temperature and hydration, thickness of subcutaneous adipose tissue, ethnicity, age, gender and mental state, as well as environmental conditions such as ambient temperature and relative air humidity [359,370–372].

The deepest layer of the skin, the subcutaneous tissue or hypodermis, consists mainly of an adipose tissue, which contributes to thermal insulation, stores energy and acts as a shock absorber [335,373]. The subcutaneous adipose layer covers the entire face with the exception of the eyelids, nose and ear pinna [373]. Obese people have a thicker layer of subcutaneous fat, which is why they sweat more when they overheat [370]. The subcutaneous tissue also contains larger vessels that branch out towards the surface of the skin and supply the sweat glands, hair follicles, and subpapillary capillary networks [359]. In the subcutaneous tissue are also inserted unique muscles whose contractions allow for various facial expressions [374] which influence blood flow and temperature patterns in the face [4]. Blood flow is also altered by changes in posture, with relaxation and stillness allowing the body to adapt to the new posture and stabilise the blood flow and thus skin temperature [375–377].

2.2. Facial skin topography

Another specific feature of the face is its varied topography, which consists of a number of rounded surfaces positioned at different distances from the observer. For this review, we have divided the front and side view of the face into 6 regions (Fig. 4): Forehead, eyes, nose, cheeks, chin and ear. The eyebrows and, in men, the cheeks, mouth and chin are covered with thick, stiff and pigmented terminal hair, while the rest of the face is covered with short, soft and fine vellus hair, which is usually without pigmentation [378]. There are no significant temperature differences between the left and right sides of the face, but asymmetry can occur in cases of inflammation, infection, tumours, circulatory disorders or mechanical tissue damage [175,379]. The face is also one of the few parts of the body that is practically always exposed. When measuring the skin temperature of parts of the body that are normally covered by clothing, the person being measured can feel uncomfortable, which may influence the measurement results.

2.3. Facial skin temperature dynamics

A third specific feature of the face is that the temperature of the facial skin changes faster and more significantly than the body temperature (core temperature) [337,338,380]. The circadian rhythm of facial

skin temperature follows that of body temperature [343], so facial skin is of interest in studies of normal and elevated body temperature. They usually observe the area of the inner canthus, which is consistently the warmest area of the face [11,381]. However, it is important to be aware that the relationship between inner canthus temperature and body temperature is not known, not standardised and not linear and therefore the difference between the two is not constant [11,17]. This is reflected in the inconsistencies between studies on the reliability of the inner canthus area in estimating body temperature [28,29,45,54,116,133,145,279].

3. How does a thermal imaging camera measure skin temperature?

The operating principle of the thermal imaging camera is based on the measurement of the infrared radiation emitted by the surface, which is converted into a digital map of the surface temperature [382]. Every surface whose temperature is higher than 0 K emits electromagnetic waves into the environment and transfers heat in this way. The properties of the surface influence the directional and spectral distribution of the radiation flux. The ideal surface can be regarded as a black body that absorbs all the incident radiation. The spectral distribution of the radiation flux density of the black body at temperature T and wavelength λ is determined by Planck's radiation law

$$L_{\lambda,b}(\lambda, T) = \frac{2hc^2}{\lambda^5} \left[e^{\frac{hc}{\lambda kT}} - 1 \right]^{-1} = \frac{1}{\pi} \frac{dW}{d\lambda} \quad (1)$$

where h is Planck's constant, k is Boltzmann's constant and c is the speed of light in a vacuum. The total energy flux density (total emissive power) W can be determined by integration of the spectral distribution $L_{\lambda,b}$ over all wavelengths of the electromagnetic waves λ , which is described by the Stefan–Boltzmann law

$$W = \pi \int_0^\infty L_{\lambda,b}(\lambda, T) d\lambda = \sigma T^4 \quad (2)$$

where σ is the Stefan–Boltzmann constant and T is the temperature of the black body. Thermal imaging cameras use the Stefan–Boltzmann law to convert the measured radiation flux density into the temperature of the observed surface.

A thermal imaging camera does not directly measure the temperature of the observed surface (e.g. human skin), as the sensor measures the entire environment within the field of view (FOV) [382,383]. The detector receives the radiation from the surface of the target object and the radiation from the surroundings that is reflected via the surface of the target object, both of which are attenuated when passing through the atmosphere [382] (Figs. 5 and 6). According to Kirchhoff's law, since the atmosphere absorbs part of the radiation it will also radiate it. The total radiation power received by the camera detector W is therefore the sum of the emissivity from the surface of the object, the reflected emission from the ambient source and the emission from the atmosphere [382] (Figs. 5 and 6)

$$W = \varepsilon \tau W_{obj} + (1 - \varepsilon) \tau W_{amb} + (1 - \tau) W_{atm} \quad (3)$$

where ε is the emissivity of the surface of the target object, τ is the transmission through the atmosphere, W_{amb} is the emitted energy from the object surroundings, $(1 - \varepsilon)$ is the reflectivity of the surface of the target object, W_{atm} is the emitted energy from the atmosphere, and $(1 - \tau)$ is the emissivity of the atmosphere (Fig. 5). When carrying out measurements with a thermal imaging camera, it is therefore necessary to properly evaluate the properties of the observed surface and its surroundings. High-quality thermal imaging cameras usually require manual setting of the value of the emissivity coefficient of the observed object, the object distance, the reflected temperature, the ambient temperature and the relative humidity.

The response curve of the detector must be matched to the range of infrared wavelengths that pass through the atmosphere with little attenuation, also known as the atmospheric window [382]. One window

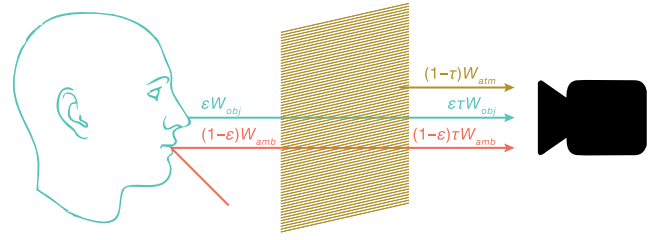


Fig. 5. Radiation power of the human face received by the camera detector (Eq. (3)).

is in the range of 2–5.6 μm , known as the medium wavelength infrared (MWIR) band, and the other in the range of 8–15 μm , known as the long wavelength infrared (LWIR) band [382]. The approximate range of facial skin surface temperature in the literature reviewed ranges from 21.4 $^{\circ}\text{C}$ to 39.0 $^{\circ}\text{C}$, so thermal imaging cameras in facial thermography typically observe the LWIR range [382,383]. The fact that LWIR cameras are more suitable for measuring facial skin temperature than MWIR cameras is confirmed by Wien's displacement law

$$\lambda_{peak} = \frac{b}{T} \quad (4)$$

where λ_{peak} is the wavelength at which the emitted radiation reaches its peak value, b is Wien's displacement constant (2898 $\mu\text{m K}$) and T is the absolute temperature of the observed skin surface. According to Wien's displacement law, the strongest emitted radiation of the facial skin is at wavelengths of 9.3 μm (39.0 $^{\circ}\text{C}$) to 9.8 μm (21.4 $^{\circ}\text{C}$), i.e. in the LWIR range.

3.1. Facial skin emissivity

The first term in Eq. (3) states that it is necessary to evaluate the radiation of the surface. For this reason, we must give the camera the appropriate value for the emissivity coefficient ε of the observed human skin. Since real surfaces are grey bodies that radiate less than black bodies, the emissivity coefficient must be taken into account, which determines the ratio between the radiation of a grey body and a black body at the same temperature and under the same spectral and directional conditions. Under ideal conditions, when the contributions of the environment and the atmosphere are negligible, the camera should only correct the measurement by the emissivity coefficient (Eq. (3)). Taking the emissivity coefficient ε into account, the Stefan–Boltzmann law is equal to

$$W = \varepsilon \sigma T_{obj}^4, \quad 0 \leq \varepsilon \leq 1 \quad (5)$$

The coefficient $\varepsilon = 1$ indicates a black body with complete absorption, and $\varepsilon = 0$ indicates a white body with zero absorption. Human skin absorbs radiation well in the infrared spectrum and is similar to a black body in this respect [379], so its emissivity factor is also very high, almost 1 [384,385]. The reason for this high value probably lies in the rough surface of the skin and the high water content of the skin [386], as described in Section 2.1. The general default value for dry clean skin is 0.98 [1,17,18,339,383,387,388], although studies consider other values such as 0.95 [29], 0.96 [129,240], 0.97 [36,123,165,177,295], 0.99 [100] and 1 [56,320]. The inconsistency of skin emissivity values is also evident in a number of studies that have measured skin emissivity *in vivo* in MWIR and LWIR spectral range, as shown in Table 2.

Togawa et al. [391,395] assume that these discrepancies in the reported values of the skin emissivity coefficient are due to the measurement methods used and also suggest that the emissivity of the facial skin is not uniform. The reasons for these differences may also be hidden in the anatomy of the skin (see Section 2.1), as many factors influence SC hydration and the size of facial pores and thus the smoothness of the facial surface, skin blood flow and the resulting

Table 2Studies on the *in vivo* measurement of skin emissivity. If they measured the emissivity on more than one skin area, the value given in the table refers to the facial area.

Study	Year	Skin area	Method	Spectral range	Skin emissivity
Watmough et al. [389]	1968		Infrared scanner	2.0 – 5.4 μm	$0.98 \leq \epsilon < 1$
Steketee [390]	1973	Forehead	Monochromator	3 – 14 μm	0.98 ± 0.01
Togawa [391]	1989	Forehead	Reflectance measurement using step change in ambient temperature	8 – 14 μm	0.969 ± 0.004
Boylan et al. [392]	1992		Reflectance measurement using step change in ambient temperature	8 – 14 μm	0.961 ± 0.007
Sanchez-Marin et al. [385]	2009	Back of the palm	CO ₂ laser beam	10.6 μm	0.990 – 0.999
Villaseñor-Mora et al. [393]	2009	Back of the palm	CO ₂ laser beam	10.6 μm	0.976 ± 0.006
			Direct method using theoretical black body	10.6 μm	0.978 ± 0.008
Charlton et al. [394]	2020	Inner upper arm	Tape with known emissivity	7.5 – 13.0 μm	0.96 – 0.99

temperature gradient of the skin, and the activity of pilosebaceous glands and eccrine sweat glands. It would be interesting to investigate the effect of sebum excretion and roughness of the skin surface on the emissivity coefficient, as even under normal atmospheric conditions (temperature 20–22 °C, relative humidity 40–60 %) people can have a large amount of sebum excretion and high skin hydration on their forehead [396]. The amount of sebum and hydration are also affected by different skin types (from dry to oily) [366]. In the case of measuring the temperature of the skin, which is warmer than the environment, and the considered emissivity is higher than the true emissivity, the camera will incorrectly predict some of the incipient radiation in emitted rather than reflected radiation and the measured temperature of the observed surface will be underestimated [383]. For typical values of the emissivity coefficient (0.98 ± 0.02), this error is about ± 0.5 °C [383,389,397]. In the LWIR and MWIR spectral range, skin pigmentation and gender probably have no significant influence on the value of the emissivity coefficient [385,386,390,391,394], while differences may occur in skin treated topically with an ointment, gel or solution [388].

With an emissivity coefficient of 0.98, a total hemispherical emissivity of the skin surface is assumed that takes into account all wavelengths and (hemispherical) directions. Table 2 shows that the emissivity is very high in both the MWIR and LWIR range and is similar to that of a black body. Less is known about the influence of the angle of incidence on emissivity, also known as directional emissivity. Watmough et al. [398] have theoretically investigated the dependence of emissivity on the angle of incidence of the surface. They came to the conclusion that the temperature measurements of the surface with an emissivity coefficient close to 1 are almost independent of the angle of incidence up to a value of 45°. At an angle of incidence of 50°, the error is about 0.5 °C, which increases rapidly with increasing angle. Vardasca et al. [189] observed an error of 0.5 °C at a smaller angle of incidence of 15°. To minimise the error, the skin surface should therefore be observed at an angle that is as close as possible to the normal angle of incidence ($\phi = 0^\circ$ in Fig. 9).

3.2. Environmental conditions

The second and third terms in Eq. (3) describe the influence of surrounding radiation reflected via the object surface and atmospheric radiation on the measurement. To reduce reflected radiation, it is necessary to avoid or remove potential sources of radiation such as heating ducts, windows, light fixtures, direct or indirect sunlight and hot water pipes in the surroundings of the observed person [17,384]. Choosing the right background is also important. In a laboratory environment, it is typically assumed that the background temperature is the same as the air temperature [383], since this type of environment allows better control over the properties of the background, which should be made of non-reflective material [17,384]. It can be seen from the second term

in Eq. (3) that the error in the measurement of absolute temperature due to background reflectance is proportional to the reflectivity of the object ($1 - \epsilon$).

Transmission through the atmosphere τ is most affected by water vapour, carbon dioxide and ozone [383,399]. The influence of these atmospheric gases increases with distance [383]. When measuring the temperature of the face, the distance between the camera and the person is relatively small (Table 4), but the relative humidity (RH) will still evidently affect the measurement, so it must be controlled [383]. RH reflects the percentage of moisture the air can hold at a given temperature. According to the third term in Eq. (3) the error in the measurement of absolute temperature due to atmospheric radiation is proportional to the emissivity of the atmosphere ($1 - \tau$). In the LWIR measurement, which is carried out at an ambient temperature (AT) of 25 °C and a RH of between 30 % and 70 %, the error is less than 1 % [399]. Larger errors are noticeable in the MWIR range [399], to which users of this type of camera should pay particular attention.

4. How to accurately measure the temperature of the face with a thermal imaging camera?

According to the previous sections, measuring facial skin temperature with a thermal imaging camera presents a unique challenge, as many factors must be taken into account in order to ensure accurate measurements. There are three main focus points when designing and executing thermal imaging of the face: (1) thermal imager, (2) subject, and (3) ambient conditions.

4.1. Thermal imager

An essential component of a proper thermographic measurement is a suitable thermal imaging camera. When choosing the type and model of a thermal imaging camera, the decision is mainly based on the clearly defined research objectives and the available budget, which lead to the selection of a suitable camera. The key technical specifications to consider are the spectral range, thermal sensitivity and image resolution.

As explained in Section 3, MWIR and LWIR thermal imaging cameras operate in wavelength ranges where infrared radiation passes through the atmosphere with minimal absorption, with LWIR being the more suitable option for temperature measurement in the facial skin temperature range. The majority of the papers reviewed, 83.0 %, follow this recommendation and use an LWIR camera. Even though MWIR cameras are better suited to measure higher temperatures than those of the facial skin, the enhanced sensitivity of MWIR cameras with a cooled detector would be an advantage for research work in the field of thermal imaging of the face.

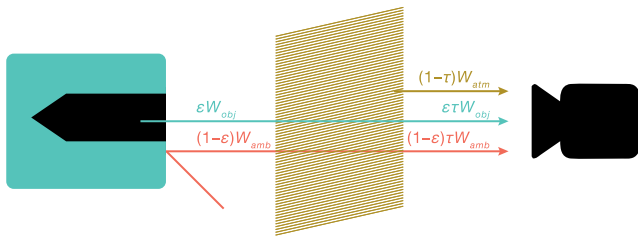


Fig. 6. Radiation power of the black body radiator received by the camera detector (Eq. (3)).

Thermal sensitivity represents the minimum temperature change that the camera can detect and is often expressed as the Noise Equivalent Temperature Difference (NETD), with a lower NETD value indicating a higher thermal sensitivity. In practise, this means that a camera with a NETD of 30 mK can detect a temperature difference of 0.03 °C between any two pixels in image. The number of pixels in image is indicated by the resolution, where a higher number of pixels means a higher resolution, which enables better differentiation of objects, finer detail recognition and more accurate temperature measurements. In reviewed papers, the camera resolution varied from 80 x 60 pixels to 1024 x 1024 pixels. Higher thermal sensitivity and resolution increase the cost of a camera. Since some thermal imaging camera models allow interchangeable optical lenses, it should be noted that the resolution and temperature sensitivity are tied to the camera detector, while the lens determines the FOV, i.e. the extent of the observable area that the camera can see at a given time. A narrower FOV, for example, condenses the pixels into a smaller area and thus enables a more spatially detailed measurement of the temperature in this area. The lens also influences the focal length, which determines the distance for optimal measurement of the surface temperature. For example, a lens with a short focal length can capture a larger scene from a shorter distance.

4.1.1. Thermal imager calibration

In order to establish metrological traceability of the thermal imaging camera to primary standard, it must be calibrated in an (accredited) laboratory or calibrated in-house by comparison with a black body reference radiator. A black body radiator is a cavity with controlled surface temperature, inside walls covered with material with known emissivity and an opening through which a thermal imaging camera can observe and measure the temperature inside the cavity (Fig. 6). An excellent black body radiator has an emissivity coefficient greater than 0.999 [17]. When calibrating a thermal imaging camera with a black body radiator, the camera must be positioned at an appropriate distance where the field of view is smaller than the opening of the cavity and the camera must be pointed directly at the centre of a black body radiator, as shown in Fig. 6. If the emissivity factor of the camera is set to 1, there should be no discrepancy between the known temperature of the black body radiator and the temperature measured by the camera. The calibration therefore specifies the measurement error for each pixel. Calibration institutes normally issue calibration certificates, which are usually based on a black body calibration.

4.1.2. Emissivity coefficient

One of the parameters that must be assigned to the thermal imaging camera to enable accurate measurement of facial skin temperature is the appropriate value for the emissivity coefficient. Different areas of the face may have slightly different emissivity coefficients due to apparent variations in facial skin type, texture and other characteristics, which can lead to inaccurate and inconsistent temperature measurements (Sections 2 and 3.1). Despite the fact that the emissivity coefficient ϵ is an extremely important parameter for the accuracy of thermal imaging, only 31.4 % of the papers reviewed reported the ϵ

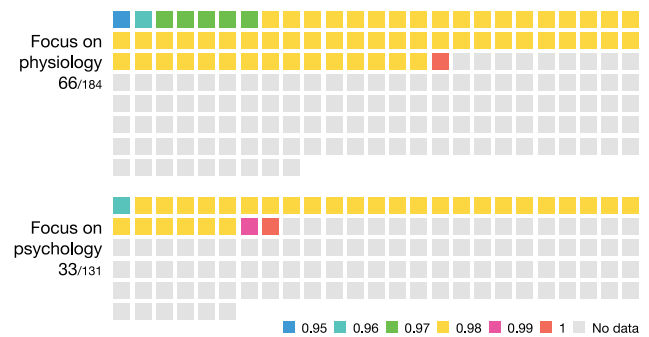


Fig. 7. Skin emissivity coefficient values as listed in the reviewed papers, categorised according to the focus of their topic.

Table 3

Skin emissivity coefficient values as listed in the reviewed papers, categorised according to the focus of their topic.

Emissivity ϵ	Focus on physiology	Focus on psychology
0.95	[29]	
0.96	[129]	[240]
0.97	[36,123,165,177, 295]	
0.98	[21,22,24,27,28,31, 32,34,38–40,42,44, 45,48,50,54,55,65, 66,70,71,73,111–113,115–118,125, 126,131,138–141,145,148,150, 152,156,159,169, 175,176,179,181, 187,188,191,277, 279,283,285,291, 314,318]	[74,75,84,86,88–91, 93,94,99,105–107, 109,197,205,208, 224,231,234,235, 244,260,262,265, 270,301,304,305]
0.99		[100]
1.00	[56]	[320]

value they used (Table 3 and Fig. 7). Although it is still a matter of research, the majority, 88.9 %, of the reviewed papers, which stated the ϵ , follow the recommendations [17,18] and use a value of 0.98 for the emissivity coefficient, regardless of which part of the facial skin they measured. It should be noted that this value applies to clean, dry skin that has not been exposed to the sun, cosmetic procedures or heavy physical activity in the last few hours. Two papers [136,294] correctly assumed an emissivity coefficient in the range of 0.98–0.99, but did not specify which value they used.

4.1.3. Measurement distance and angle of incidence

The camera and lens specifications affect the measurement distance at which the camera can accurately measure the temperature of the observed surface. In more than half of the papers reviewed, 68.9 %, the measurement distance was specified (Table 4 and Fig. 8), with the majority, 88.5 %, carrying out the measurement at a distance of 2 m or less. Shorter distances allow the use of less expensive thermal imaging cameras with lower resolution. The appropriate measurement distance is as close as possible to the minimum focus distance [17], which is normally specified in the product manual. However, if the application requires a greater measurement distance, this should be within the distance range specified in the manual of the camera used. It is important that the entire face or a part of the face of interest is within the thermal image at the selected measurement distance. Appendix B contains a formula for the informative calculation of the appropriate measurement distance.

In addition to the measurement distance, the angle of incidence at which the camera measures the surface temperature of the facial skin is

Table 4
Measurement distance d values as listed in the reviewed papers, categorised according to the focus of their topic.

Distance d	Focus on physiology	Focus on psychology
$d \leq 1$ m	[20,21,23–25,28–32,35–37,40–44,47–49,52–57,60,61,63–69,71–73,111–113,115–120,122–126,131,134–136,138–141,143,145,147,149–152,154–161,164–167,169–173,175,183,188,189,191,192,283,292,295,317,318,330]	[74–76,78,81,84,85,87,88,90,93–95,97–99,101–106,109,194,195,197,201,203,204,210,215,216,218,220–222,224,228,229,231,235,236,240,241,244–248,250,251,254,257,258,260,263,265–267,275,304,305,319,333]
$1\text{ m} < d \leq 2\text{ m}$	[21,27,34,38,39,45,52,70,73,110,114,129,133,134,148,161–163,170,173,177–181,185,189–191,276,277,279,288,290,291,297,299,314,318]	[86,89,91,100,200,208,209,212,243,252,261,262,272,275,301,303,309,333]
$2\text{ m} < d$	[21,26,73,128,130,134,161,173,191,280,281,293]	[80,107,231,255,269,273]

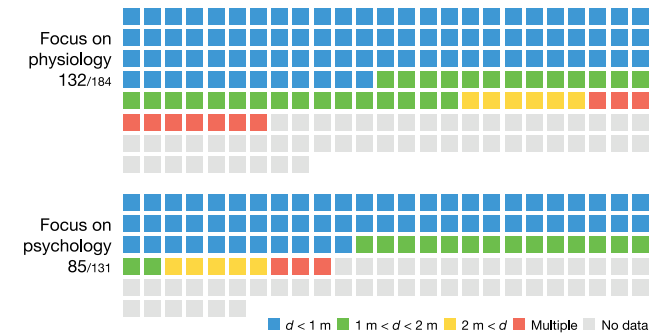


Fig. 8. Measurement distance d values as listed in the reviewed papers, categorised according to the focus of their topic.

also important for the positioning of the thermal imaging camera. For accurate and consistent measurements, the camera should be positioned perpendicular to the face (normal incidence), as shown in Fig. 9, to avoid variations in the emissivity coefficient, to ensure that the FOV is uniformly focused, to avoid possible geometric distortions and to ensure consistency of repeated measurements (even between different individuals). However, contours, curves and varying angles of the face can lead to inaccurate temperature measurements as it is virtually impossible to maintain a normal incidence on every part of the face. This can lead not only to a varying emissivity but also to a varying focus on the face, especially because some parts of the face are closer to the camera than others. This can result in some parts of the face being out of focus, affecting the resolution and clarity of the thermal data in these areas. In practise, this means that in a thermogram of the front of the face, the temperature of the cheeks near the ear may be lower due to the curved surface. This makes it very difficult to interpret the results correctly, e.g. whether the measured cheek temperature is lower due to the surface curvature and poorer focus or for psychological reasons. If the experimental conditions do not allow measurement at normal incidence, the angle of incidence should be kept as constant as possible to ensure a reliable measurement.

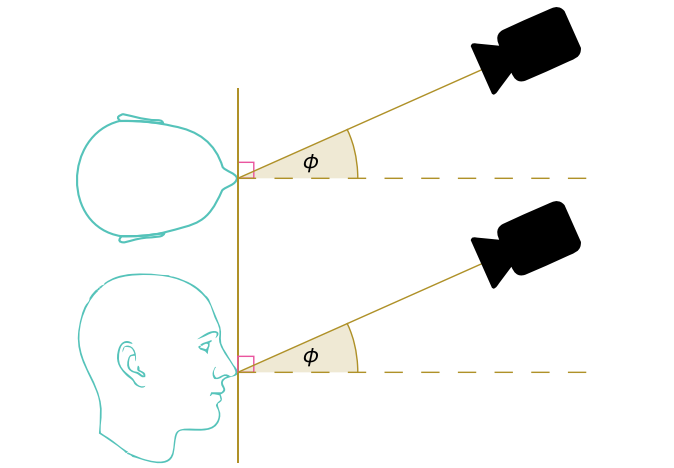


Fig. 9. Angle of incidence ϕ of the thermal imaging camera. The camera should be positioned as close as possible to the normal incidence (dashed line).

4.1.4. Thermal imager acclimatisation

To achieve accurate measurements, the camera should also be thermally stabilised so that the detector can warm up and reach a stable operating temperature, which improves the NETD. Switching the camera on for an appropriate period of time before starting the measurement will allow the camera to reach thermal equilibrium with the environment, minimising any possible thermal drift in the measurements due to expansion and contraction of the camera components. The time required for the camera to stabilise depends on the camera model and is usually specified in the manual. For the latest models, this time is around 5–10 min, with higher quality and more expensive cameras requiring less time. Starting the examination with a thermally stabilised camera also helps to ensure that the measurements are comparable between subjects or across multiple examinations of the same subject.

4.2. Subject

The accuracy of measuring facial skin temperature with a thermal imaging camera also depends on adequate preparation of the subject, starting a few hours before the measurement begins. If this is not the purpose of the research, the subject should avoid all external factors that may cause undesirable changes in facial skin temperature, such as the consumption of alcohol, caffeine and large meals, the application of cosmetics and make-up and other factors such as smoking, heavy physical activity and sunbathing. TISEM [18] recommends avoiding these conditions for at least four hours before the measurement, which the subject should confirm verbally before starting the measurement.

4.2.1. Skin emissivity coefficient

An emissivity coefficient of 0.98 is recommended for dry clean skin (Section 3.1). Therefore, not only should you not apply cosmetics or make-up, but the skin must also be free of sweat, sebum and other impurities. Nowadays, it is not uncommon to use facial cosmetics such as moisturisers and sunscreens. Even if these were applied long before the measurement and the effect of surface cooling by evaporation can be excluded, there is still the possibility that the applied cosmetics influence the emissivity of the observed surface [388]. Thus, it is also problematic if the cosmetics leave a visible greasy film on the skin or if make-up is applied, as this changes the properties of the surface observed by the thermal camera. For similar reasons, it is also problematic to observe oily skin without cleansed sebum. If necessary, the skin should be dried or cleansed before the measurement, which should be clearly described in the measurement report [17,18].

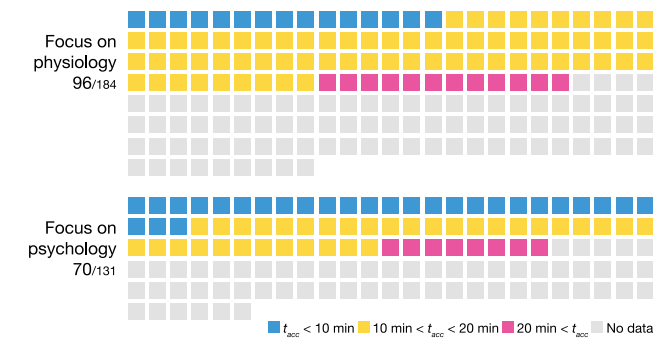


Fig. 10. Length of subject acclimatisation period t_{acc} values as listed in the reviewed papers, categorised according to the focus of their topic.

Table 5

Length of subject acclimatisation period t_{acc} values as listed in the reviewed papers, categorised according to the focus of their topic.

Acclimatisation period t_{acc}	Focus on physiology	Focus on psychology
$t_{acc} \leq 10$ min	[28,61,70,120,128,138,140,161,175,180,181,189,281,324,330]	[74,81,84,85,88,90,91,93,97,102,105,195,214,221,229,236,242,246–248,254,263,264,270,301,306,319,334]
$10 \text{ min} < t_{acc} \leq 20$ min	[22,24,25,27,29,31,33,34,36–38,40,44,50,52–56,65,68,71,72,112–114,118,119,121,122,135,139,141,142,145,148,150–152,157,159,176,177,179,182,183,187,188,191,283,287,293,295,296,300,310–316,321–323,325–327,329]	[75–77,80,82,86,94,98,101,107,108,201,203,204,208–213,218,220,224,227,228,232,234,237,240,250,253,261,266,304]
$20 \text{ min} \leq t_{acc}$	[21,39,46,47,111,116,129,146,147,282,290,317]	[95,99,109,197,207,251,257,258]

4.2.2. Subject acclimatisation

Another important part of the preparation of the subject is an adequate acclimatisation period so that the facial skin can achieve thermal equilibrium with the environment. In this way, the facial skin can reflect the internal physiological conditions without being influenced by external factors. The length of the acclimatisation period depends on the research protocol and the initial difference between the external environment and the temperature in the test room, but should be at least 10 min [17]. The acclimatisation period was reported in almost half of the reviewed papers, namely in 49.4 % (Table 5 and Fig. 10). The duration varied between 2 and 60 min, with a median value of 15 min. During the acclimatisation period, the subject usually sits in a comfortable position in a quiet environment, according to the reviewed papers. In some studies, this time is used to conduct a calm conversation in which the measurements are explained and any questions the subject may have are answered. In psychological research, it is also commonly used to reduce the level of arousal and achieve a calm and relaxed mental state (see Fig. 10) (see Table 5).

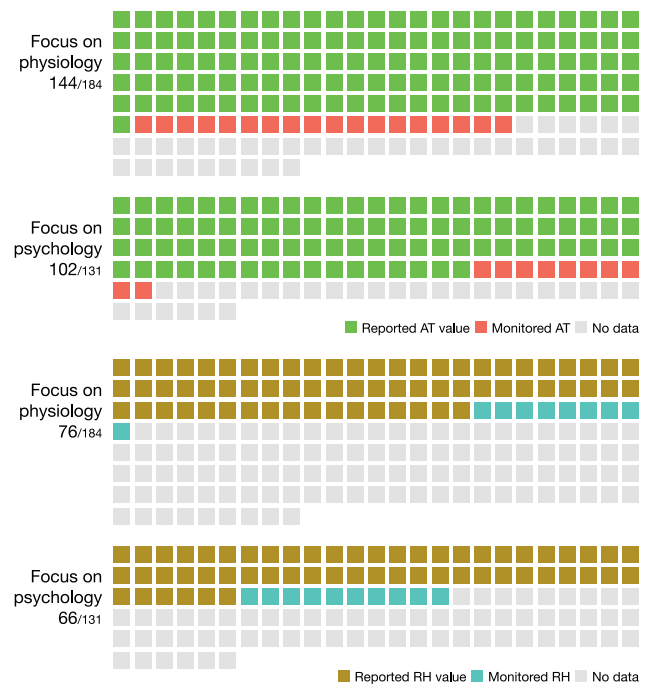


Fig. 11. AT and RH values as listed in the reviewed papers, categorised according to the focus of their topic.

4.3. Ambient conditions

Both the camera and the subject are sensitive to the ambient conditions. Ambient temperature (AT) and relative humidity (RH) affect both the skin temperature (Section 2) and the radiation measured by the camera detector (Section 3), so the skin temperature should be measured in a controlled environment whenever possible. This means that the ambient conditions should not only be recorded, but should also be relatively stable [1,17,18]. Unstable ambient conditions also influence the accuracy of the measurement and contribute to measurement uncertainty, as the AT and RH values set in the thermal imager deviate from their actual values [17]. Therefore, shaded windows and a suitable air conditioning and ventilation system are recommended, whereby the air flows must not be directed directly at the observed subject. The studies reviewed are aware of this, with 75.1 % of studies reporting AT and 41.5 % reporting RH (Table 6 and Fig. 11). It is recommended that the AT value is not below 18 °C and not above 24 °C so that the clothed person feels comfortable and no thermoregulation processes are triggered (sweating or shivering) [1,10,12,17]. Exceptions are various provocation tests and physiological research that examine such processes and require a lower or higher AT value [1]. Taking into account the anatomy of the skin and the performance of the thermal imaging camera at the recommended AT value, the recommended RH value is between 30 % and 70 % [17,354–356,399].

As described in Section 3.2, the thermal imaging measurements are also influenced by radiation sources in the FOV and in the vicinity of a subject as well as by reflective background surfaces (such as glass, mirrors and metal). Therefore, it is necessary to perform the measurement in a room without radiation sources (except the human subject) and with a non-reflective background in the FOV of the thermal imager. The experimental room should also be spacious enough to ensure adequate distance between the subject, thermal imager and the walls.

5. How to analyse thermograms?

The result of a thermal imaging measurement is a temperature image, or thermogram, whose pixels contain temperature information.

Table 6
AT and RH values as listed in the reviewed papers, categorised according to the focus of their topic.

Parameter		Focus on physiology	Focus on psychology
AT	Reported value	[20–29,31–43,46–51,53–56,59,62,64,65,69–73,111–116,118,120,121,123,125,126,128–131,133,134,137,139–143,145–152,154–160,163,168–171,173,176–181,183,186–191,276,277,282,283,289,290,292,295,296,300,310–318,321,322,327–330,332]	[74–78,81–91,93–95,97,100–109,194,195,199,201,203,204,206–211,213,215,216,219–223,227–237,240,242,243,246–248,250–254,257,258,261–267,270,272,274,303,305,306,308,309,320,333,334]
	Monitored AT	[52,61,110,122,135,136,138,162,164,166,174,175,182,280,284,293,297,325]	[80,99,212,218,238,241,245,260,304,319]
RH	Reported value	[21–25,27–29,34,37–41,43,46,48–50,53–55,64,65,69,70,72,73,114,116,118,121,123,125,128–131,133,139,140,142,145,147–151,154–158,169,177,179,181,183,189,191,282,283,290,296,314,317,318]	[74–77,81–86,88–91,94,100–103,105–108,199,203,204,206–211,213,215,216,227–229,235–237,240,242,243,246–248,250,252–254,263,265,266,270,303]
	Monitored RH	[36,52,122,134,168,188,297,300,312]	[87,97,212,218,238,241,245,260,274,304]

Different temperatures are represented on the thermogram by different colours, whereby the user can select the desired colour palette. The accuracy of the thermal imaging measurement is also influenced by the way in which the thermograms are analysed. The basis of thermogram analysis is the selection of the region of interest (ROI), although the less common terms area of interest (AOI) and point of interest (POI) are also used in the papers. The ROI comprises the pixels of the thermogram that cover the region of interest, which may include the entire face or only a specific part (Fig. 12). The selection of the ROI can be done manually by drawing the desired shape on the thermogram, e.g. a circular ROI at the tip of the nose, or automatically using algorithms and machine learning tools. The latter enables faster processing of thermograms, so that it is also possible to process videos and observe the temporal dynamics of temperature changes in the facial skin.

5.1. Size, shape and placement of ROI

There are no standardised rules for defining the size, shape and placement of the ROI. It is however recommended that the ROI covers as much of the field of view as possible or contains as many pixels as possible. According to recent research [400], the ROI should be at least 3 × 3 pixels in size and surrounded by a band at least 7 pixels wide so that the average temperature of the ROI can be calculated correctly due to the size-of-source effect (SSE) (Fig. 12). Only 19.4 % of the papers in the review explicitly stated the size of the ROIs, however the majority of these papers (82.0 %) had a ROI size of at least 3 × 3 pixels.

The categorisation of the papers in this review article into physiologically oriented and psychologically oriented research is mainly based on the approach to defining ROIs. In physiologically orientated research, ROIs are placed in areas that are pathologically significant,

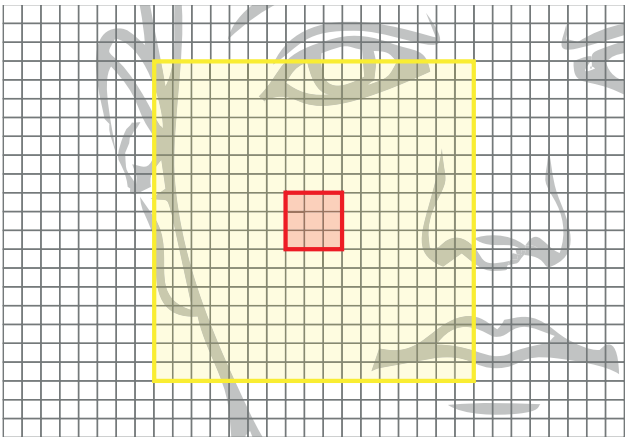


Fig. 12. Minimum size of the ROI (red area) for an accurate measurement of the average temperature within the ROI, surrounded by a band of 7 pixels (yellow area).

e.g. measuring forehead temperature to investigate migraine. Also, the shape of the ROI is usually tailored to the anatomical features to ensure that the desired regions of interest are covered as faithfully as possible. Most of these studies placed the ROI on the forehead (39.7 %) and cheek (38.0 %) (Fig. 7 and Table 7). Interestingly, 29.9 % of the papers focusing on physiology placed the ROI over the entire front or side of the face. Psychological studies, on the other hand, usually examine the areas of the face in which the greatest changes are to be expected as a result of the psychological variable being studied. In the psychologically orientated literature the ROI is most frequently placed on the nose (69.5 %) and forehead (61.1 %) (Fig. 13 and Table 7).

Although various studies state that they observe ROIs on the same part of the face, the shape and placement of these ROIs may differ from each other. Fig. 14 shows the most common ROI shapes and placements for specific parts of the face. The figure shows that the ROIs have a fairly simple shape, but are very difficult to identify due to the complex shapes of the different parts of the face, which can vary greatly from person to person. To ensure the repeatability of manually selected ROIs, two or more independent raters were used in some studies [80,81,90,112,119,128,148,212,228,246,247,295].

5.2. Data analysis within a ROI

Once the ROI has been determined, the selected pixels are statistically analysed. Usually the mean temperature and the standard deviation are calculated, but often also the minimum and maximum temperature within the ROI to estimate the temperature distribution within the ROI. The average value of all pixels within the ROI is problematic if some pixels cover areas with terminal hair, nostrils, open mouth or skin anomalies, which affects the average temperature of the entire ROI (Fig. 15). To improve the robustness of the average value, some studies include only a certain percentage of the warmest pixels within the ROI in the calculation, so that the average value is less sensitive to the values of the pixels covering the anomalous areas. This percentage varies between 5 [39] and 50 [304], with the most common value being 10 % [196,200,260,308]. Robinson et al. [260] assessed variation and statistical power using cut-offs of 5 %, 10 %, 20 %, 30 %, 40 % and 50 % and found that 5 % was too restrictive, while 40 % and 50 % were not restrictive enough. As the difference between the remaining percentages was very small, they concluded that the best option was to calculate the mean of 10 % of the warmest pixels within the ROI. However, this type of statistic should also fulfil the condition of the minimum number of pixels taken into account in the calculation. The calculated statistics are then used for analysis depending on the purpose of the study. The most straightforward analysis is to observe

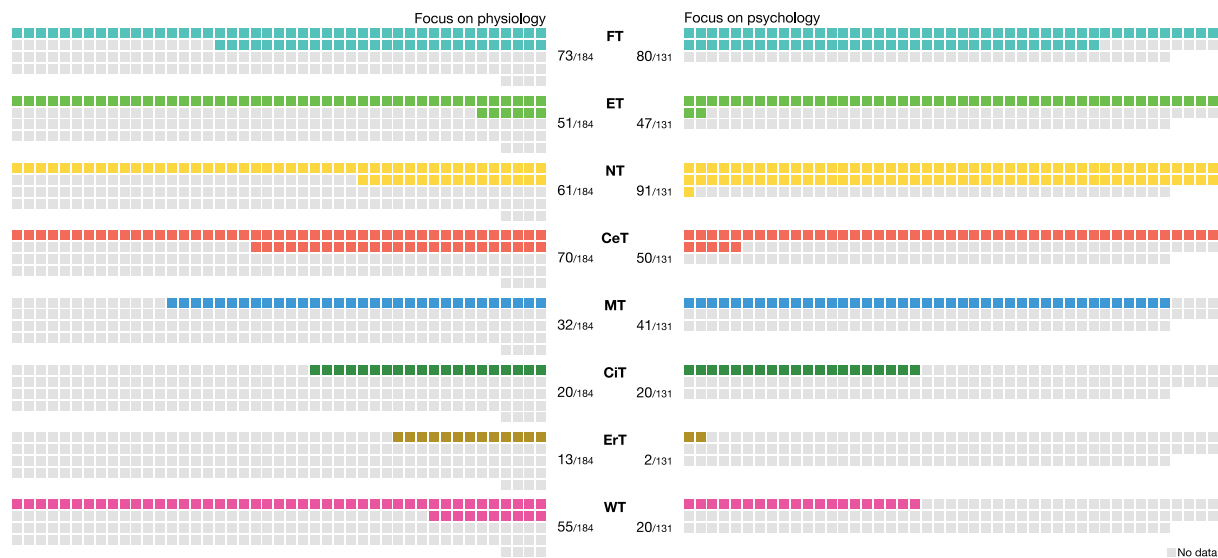


Fig. 13. ROI overview with parts of face labelled FT—forehead temperature, ET—eye temperature, NT—nose temperature, CeT—cheek temperature, MT—mouth temperature, CiT—chin temperature, WT—face temperature, ErT—ear temperature.

Table 7
Overview of selected ROI in reviewed literature.

Facial part	Focus on physiology	Focus on psychology
Forehead	[21,24,25,27,31,32,36,38,39,43–45,47,51,54,56,68–70,73,112,113,115,118,119,121,127,129,130,133–135,139,140,145,146,148–150,153,157,160,176–178,180,182,183,186,276,277,279,281–283,287,291,292,295,299,311,314,316,318,321–323,327–332]	[74–79,81,83–91,94,95,98,100,102–104,106–108,194–196,199,201,204,206–210,212,214,217,220,221,224–226,228–231,235,236,238–241,243–247,249,252,260,261,263,267,270,272,275,301–304,306–308,319,320,333,334]
Eyes	[22,24,27–29,31,32,38,39,41,43,45,49,51,54,69,70,116–121,123,125,133,140,145,149,153,157,176,180,182,183,189–191,279,281,283,288,291,292,311,321–324,327,329]	[75,78,86,90,93,95,104,196,200,201,206,207,214,218–224,228,229,233,235,238,239,241,243,246,247,250–252,254,257–263,271,274,304,307,319,320]
Nose	[21,27,32,35,38,39,41,56–58,60–65,68,70,73,114,118,119,121,129–131,134,137,144,147,149,156–158,160–164,166–174,176,180,182,276,277,283,290,292,297,298,317,321,323]	[74–92,94,95,98,101–105,107,108,194–196,199,201,203–205,207–217,220,221,224,226–231,234–253,255,261,263,264,266,269,270,272,275,303–306,320]
Cheek	[21,23,25,27,32,33,37–39,44,47,48,50,55,69,70,73,111–114,119–122,124,130,131,134,135,138,139,141,144,146–148,150,151,154,156,157,160,163,175,177,178,180,186,188,277,282,287–289,292,293,295,310,311,314–316,318,321–323,326,327,329]	[74,75,77,78,81,85–91,94,98,103,104,195,196,201,206–210,220,221,224,226,228–231,236,238,240,243,245–247,250,252,260,263,266,270,272,303,309,319,320]
Mouth	[31,32,38,39,41,44,48,51,54,65,69,119,121,134,145,147–149,156,162,167,173,174,180,183,276,277,281,282,285,288,292]	[75,78,84,90,95,98,103,198,201,203,207,208,213,214,217,220,221,224–226,228–230,233,235,236,238,239,241,243,245–247,253,256,260,261,263,265,269,320]
Chin	[21,25,27,36,69,70,73,119,120,130,131,146,157,163,176,177,180,186,282,311]	[74,77,87,89–91,95,103,108,207,209,214,220,221,228,229,243,261,272,320]
Ear	[31,45,110,118,126,130,134,183,188,277,293,315,318]	[87,236]
Whole face	[20,24,26,30,34,40,42,46,48,52–54,56,59,61,66,67,71,72,128,132,134,136,142,143,145,152,155,157,159,165,167,176,179,181,182,184,185,187,190–193,277,278,280,284,286,294,296,300,312,313,325,332]	[74,89,93,96,97,99,105,108,109,196,197,202,207,228,231–233,268,273,309]

how the ROI changes over the course of the experiment. It is also possible to make comparisons, both between different ROIs on the face to investigate temperature anomalies and asymmetries, and between different times of the experiment, typically comparing the values of a selected ROI during the stimulus with the value at baseline.

6. How to report measurement data?

The researchers should record data of the measurement conditions, the settings of the thermal imaging camera and the data from the subjects. To ensure that they have recorded all important data, they can use the TISEM checklist [18], which includes subject demographics, thermal imaging camera and room settings, and details of how the

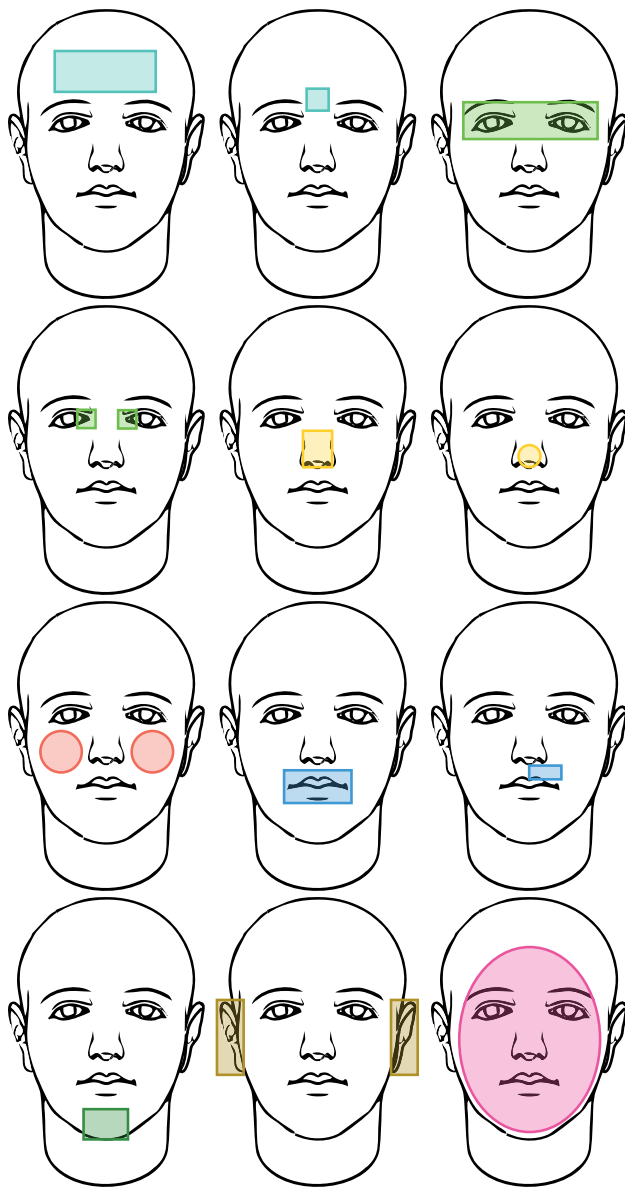


Fig. 14. Common ROI placement in reviewed papers.

thermal images were analysed. Researchers are encouraged to provide detailed information about the measurement conditions and the procedures used in thermal imaging, as this improves the interpretability of the data [18] as well as the reliability and reproducibility of the experiment.

Here are examples of reports on individual parts of facial skin temperature measurement with a thermal imaging camera. The settings of thermal imager can be described as follows: “The LWIR camera used was a FLIR T1020 (accredited calibration 5 months ago) with a resolution of 1024×768 pixels and a sensitivity of < 20 mK at 30°C . The emissivity coefficient was set to 0.98. The camera was positioned perpendicular to the face of a seated person at a distance of 1 m. The camera was switched on 5 min before the start of the measurement to ensure the stability of the sensor. The camera was mounted on a tripod that was in a fixed position.” The preparation of the subjects and their skin should also be clearly described, e.g.: “The subjects were instructed to refrain from large meals, alcohol and caffeinated drinks, smoking, sunbathing and intensive physical activity for at least 4 h before the measurement. When they arrived in the



Fig. 15. An example of an LWIR thermogram of a person, where a darker colour indicates a lower temperature. The pixels covering the beard and hair area are visibly darker and therefore these areas are colder. Glasses also appear cold as the glass reflects the infrared radiation.

room where the experiment was conducted, they remained in a seated position for 15 min to calm down and thermally acclimatise. During this time, we introduced them to the measurement and answered any questions they had. We acquired relevant personal data such as age, weight, height, gender, frequency and intensity of physical activity as well as current medication or medical treatment.”. The description of the protocol should include all intentional and accidental movements of the subject, e.g. head movements, talking during the recording, change from sitting to lying down, touching face with hands, etc. If the measurement protocol contains factors that influence skin temperature or emissivity, these should also be specified. The description of the protocol also contains the date and time of the measurement. Example for the description of ambient conditions: “The measurement was performed in a room with a stable temperature of $23.1 \pm 1.0^\circ\text{C}$ and a relative humidity of $50 \pm 10\%$. The subject was seated in front of a non-reflective background and all possible sources of radiation in the room were removed. There was no direct sunlight and no draughts.” The analysis of the thermograms can be presented as follows: “For each subject, we analysed two thermograms taken at the beginning and at the end of the task. The ROI was a circle covering the tip of the nose, as shown in Fig. 14. The size of the ROI was adjusted to each subject, with a minimum size of 120 pixels. To extract the ROI from the thermogram, we wrote a Python script that also calculated the average value of the 10 % warmest pixels within the ROI. We compared the difference between the two thermograms.”

7. What is the measurement uncertainty of measuring facial skin temperature with a thermal imager?

When assessing the accuracy and quality of the measurement performed, it is crucial to evaluate the measurement uncertainty, which is a quantitatively expressed doubt about the measured facial skin temperature [401,402]. It is expressed as a non-negative parameter that characterises the dispersion of the temperature values attributed to the measured facial skin temperature [401,402]. In the uncertainty budget for measurement of facial skin temperature, there are the following sources of uncertainty to be included: (i) accuracy, repeatability

Table 8

Uncertainty budget for measurement of black body temperature and facial skin temperature (optimal case). Best empirical estimation for uncertainty component is marked with asterisk (*).

Uncertainty component	Black body /°C	Facial skin/°C
Thermal imager	0.13 [17]	0.26 [17]
Distance	$\frac{0.1}{2\sqrt{3}}$ [17]	$\frac{0.3}{2\sqrt{3}}$ [17]
Angle	/	$\frac{0.5}{2\sqrt{3}}$ * [189,398]
Ambient conditions	/	$\frac{0.1}{2\sqrt{3}}$
Emissivity coefficient ($\epsilon = 0.98 \pm 0.02$)	/	0.5 * [383,389,397]
ROI placement and shape	/	$\frac{0.5}{2\sqrt{3}}$ *
Subject	/	$\frac{0.5}{2\sqrt{3}}$ *
Combined uncertainty	0.14	0.61
Rounded expanded uncertainty ($k = 2$)	0.3	1.3

and size-of-source effect (SSE) of the thermal imager and auxiliary instrumentation, (ii) distance and the angle of the positioning of the imager from the face, (iii) ambient temperature and relative humidity conditions during measurement, (iv) emissivity of the target object (face), (v) size, shape, positioning, manual/automatic selection of the ROI, number of pixels in the target size, (vi) influence of the protocol of measurement. In addition, errors due to reflected and background radiation, time drift of the thermal imager, non-uniformity of the skin etc. should be estimated. Table 8 shows the best uncertainty budget for a facial skin temperature measurement, performed at a typical distance and under optimal conditions and appropriately analysed as described in Sections 4 and 5, compared to the reference black body radiator, as a highly reliable and metrologically evaluated thermal source.

From Table 8 we can conclude that the measurement of the temperature of the black body can be associated with 0.3 °C, while it is much larger for the temperature of the facial skin, namely 1.6 °C. Note that this is a metrological evaluation of the measurement, which includes numerous sources of error that are not taken into account in the usual studies.

8. What have we learned?

This systematic review demonstrates the usefulness of thermal imaging of the human face for a wide range of applications in medical and psychological fields. However, it is necessary to ensure accurate and reliable measurements that allow comparability within and between studies and correct interpretation of results in order to obtain valid information about the functioning of human physiology and psychological processes. To this end, Table 9 provides basic guidelines for selecting the thermal imaging camera, preparing the facial skin of the subject, preparing the environment and analysing the recorded thermograms.

The thermal imaging camera should observe the LWIR range with the highest possible resolution and the lowest possible NETD value, as far as the budget allows. The camera settings for the emissivity coefficient should be 0.98 for dry, clean skin without cosmetics, make-up or impurities. The measurement distance should be chosen so that the face covers as much of the FOV as possible and that each ROI in the thermogram is in focus and the appropriate size, but still within the distance range specified in the camera user manual. The camera should also be placed directly in front of the face with a constant angle of incidence that is as close as possible to the normal incidence. To allow the camera detector to stabilise, the camera should be switched on for a period of time specified in the user manual before the measurement.

The preparation of the subject should ideally begin a few hours before the actual measurement. The subject should not consume alcohol, large meals or caffeine four hours before the measurement and

should not take a shower or sunbathe. There should be no visible debris, cosmetics, make-up, sweat or sebum on the facial skin. Otherwise, the facial skin should be dried or cleaned before the measurement, which should be clearly described. After the subject has confirmed that they have followed the instructions and the skin is visibly clean, an acclimatisation period of at least 15 min should be allowed. During this time, the subject should sit in the experimental room and move and exert themselves as little as possible in order to properly thermally acclimatise.

The ambient conditions should be controlled, with an AT between 18–24 °C and an RH between 30–70 %. In studies in which physiological and psychological parameters are investigated under conditions outside the specified ranges or under unstable conditions, the influence of these conditions on the accuracy of the measurement should be taken into account. There should also be no radiation sources in the FOV or in the vicinity of the subject. Therefore, windows should be covered to avoid (in)direct sunlight and other sources should be removed or covered. The background should also be made of a non-reflective material that has a different temperature than the face.

The thermogram analysis consists of defining the size, shape and placement of the ROI. The ROI must cover the part of the face that is of interest, as close to the anatomy as possible. This increases the homogeneity of the temperature within the ROI as it avoids the areas with terminal hair (eyebrows, beard, scalp hair) and various surface features. The ROI should have as many pixels as possible, but at least 3×3 pixels. This ensures an accurate measurement of the average temperature within the ROI. To reduce the variability of the temperature within the ROI, you can also calculate the average of 10 % of the warmest pixels within the ROI, whereby the condition of the minimum number of pixels is still fulfilled.

9. Conclusion

In this systematic review, an extensive corpus of references dealing with the use of thermal imaging cameras to measure skin temperature of the human face is presented and thoroughly analysed using the PRISMA method. The analysis leads to the following conclusions. For reliable, reproducible and accurate measurements of skin temperature of the human face, it is very important that the measurements are carried out with a suitable thermal imaging camera in a room with suitable conditions and that the person to be measured and the skin surface to be observed are adequately prepared. For the clarity of the results and measurement reports, it is equally important that they contain the key parameters and measurement conditions. In this paper, we propose the formulation of a relevant and necessary description of the methodology, measurement instruments and thermogram analysis to help the reader understand and recognise the study results. From the metrological point of view, an estimate of the measurement uncertainty for an artificial laboratory reference, i.e. a black body radiator, and for a physiologically dynamic surface, i.e. the human facial skin, is calculated to be 0.3 °C and 1.3 °C, respectively. For an accurate and reliable measurement of the temperature difference between experimental groups or different spatial or temporal points, the difference value should be greater than the measurement uncertainty of the black body. Smaller differences are metrologically questionable and may not contain valid thermal information about the measured area.

CRedit authorship contribution statement

Valentina Stanić: Writing – review & editing, Writing – original draft, Visualization, Validation, Project administration, Methodology, Investigation, Formal analysis, Data curation, Conceptualization. **Gregor Geršak:** Writing – review & editing, Validation, Supervision, Conceptualization.

Table 9
Proposed guidelines for thermal imaging of facial skin.

Thermal imager	
Spectral range	LWIR
Resolution	The minimum resolution of the camera must ensure the appropriate ROI size.
NETD	As low as the budget allows.
Emissivity coefficient	0.98 for dry, clean skin.
Distance	An appropriate measurement distance ensures that each ROI in the thermogram is in focus and appropriate size. The measurement distance should be within a distance range specified in the user manual of the camera.
Angle	The angle of incidence should be as constant as possible and as close as possible to the normal incidence
Acclimatisation time	The time required for the camera to stabilise is specified in the user manual. For the latest models, this time is around 5–10 min.
Subject	
Before measurement Skin preparation	No alcohol, smoking, caffeine, large meals, showering, sunbathing for 4 h before assessment. There should be no visible debris, cosmetics, sweat or sebum on the skin. If necessary, the skin should be dried or cleansed before the measurement, which should be clearly described.
Acclimatisation time	≥15 min
Ambient conditions	
AT	Controlled (18–24 °C)
RH	Controlled (30–70 %)
Background	Non-reflective material, different temperature than face.
Sources of radiation	There should be no sources of radiation in the FOV or in the vicinity of a subject, e.g. heating ducts, hot water pipes, (in)direct sunlight.
Thermogram analysis	
ROI size	Covers as many pixels within the FOV as possible, with a minimum of 3 × 3 pixels.
ROI shape	As close to the anatomical features as possible.
ROI statistics	Mean value of the pixels within the ROI. With sufficient ROI size and large thermal variability within the ROI: mean value of 10 % of the warmest pixels within the ROI.

Declaration of competing interest

The authors declare that they have no known competing financial interests or personal relationships that could have appeared to influence the work reported in this paper.

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Appendix A. Overview of reviewed studies

Table A.10 shows key information from the reviewed papers, including the emissivity coefficient ϵ , the measurement distance d_{meas} and the length of subject acclimatisation period t_{acc} . Parts of face are labelled as FT–forehead temperature, ET–eye temperature, NT–nose temperature, CeT–cheek temperature, MT–mouth temperature, CiT–chin temperature, WT–face temperature, ErT–ear temperature.

Appendix B. Measurement distance calculation

For an accurate measurement of the surface temperature of the facial skin with a thermal imaging camera, the ROI should cover at least 3×3 pixels in the FOV due to SSE (Section 5). Calculating the appropriate measurement distance requires information about the instantaneous field of view (IFOV), which indicates the size of a pixel (see Fig. B.16). The theoretically permissible maximum measurement distance d_{max} can be calculated as follows

$$d_{max} = IFOV_{mm} \cdot \frac{P}{FOV} \cdot \frac{180}{\pi} \cdot \frac{1}{1000} \quad (B.1)$$

where are $IFOV_{mm}$ camera IFOV in mm, P number of pixels in a chosen direction (horizontal or vertical) and FOV camera FOV (horizontal or vertical).

For example, we have camera with FOV $12^\circ \times 9^\circ$ and resolution 1024×768 pixels and want to measure temperature of nasal tip. We want a ROI of 3×3 pixels to cover an area of 9 mm^2 , in this case $IFOV_{mm} = 1 \text{ mm}$. Using Eq. (B.1), we can calculate the maximum measurement distance as follows

$$d_h = 1 \text{ mm} \cdot \frac{1024}{12} \cdot \frac{180}{\pi} \cdot \frac{1}{1000} = 4.9 \text{ m} \quad (B.2)$$

Table A.10

Overview of reviewed studies utilizing facial thermography.

Authors	Year	N	Topic	ROIs	Imager type	ϵ	d_{meas} [m]	AT [°C]	RH [%]	t_{acc} [min]
Abaza et al. [110]	2013	88	Human ear identification	ErT	MWIR		2	Controlled		
Abbas et al. [166]	2011	7	Respiration monitoring (neonatal)	NT	LWIR		0.7–0.8			
Abdelrahman et al. [194]	2017	12; 24	Cognitive load	FT, NT	LWIR		1	24; 26		
Abdelrahman et al. [195]	2019	22	Classifying attention types	FT, NT, CeT	LWIR		1	23		
Abdrakhmanova et al. [20]	2021	142	Multimodal dataset	WF	LWIR		1	25		
Abouelenien et al. [196]	2016	30	Deception detection	FT, ET, NT, CeT, WT	LWIR					
Abreau et al. [125]	2016	40	Dry eye syndrome	ET	LWIR	0.98	0.25	24 ± 2	40 ± 3	
Adachi et al. [197]	2019	7	Detection of drowsiness in drivers	WT	LWIR	0.98	1; 0.7			>30
Akbar et al. [198]	2019	96	Stress detection	MT	LWIR					
Aldred et al. [61]	2022	136	Respiratory rate estimation	NT	LWIR		0.8			5
Alkali et al. [169]	2017	14; 15	Respiration rate monitoring	NT	LWIR	0.98	1	24.4 ± 0.35	45 ± 0.71	
Almeida et al. [74]	2022	1	Thermal comfort evaluation	FT, NT, CeT, CiT, WT	LWIR	0.98	1	10–30	40–80	10
Antonaci et al. [115]	2018	35	Migraine diagnostics	FT	LWIR	0.98	1	23.6 ± 1.57		
Aristizabal-Tique et al. [75]	2023	3	Emotional valence and blood perfusion	FT, ET, NT, CeT, MT	LWIR	0.98	0.8	24	63	15
Ashrafi et al. [21]	2022	10	Facial thermograms database	FT, NT, CeT, CiT	LWIR	0.98	1–6.6	20.61 ± 28.01	49.96 ± 58.59	60
Aureli et al. [199]	2015	31 dyads	Still-face paradigm (infants)	FT, NT	LWIR			23 ± 1	50–55	
Baltaci et al. [200]	2016	11	Stress detection	ET	LWIR		1.15			
Bang et al. [124]	2014	15	Assessment of eye fatigue	CeT	LWIR		1			
Barbosa Pereira et al. [163]	2018	20	Cardiorespiratory signals monitoring	NT, CeT, CiT	MWIR		±1.5	22.3		
Barcat et al. [123]	2017	29	Sleep onset in preterm neonates	ET	LWIR	0.97	0.4	32.7 ± 1.4 (incubator); 25 (room)	65 (incubator); 50–60 (room)	
Batinjan et al. [122]	2014	40	Monitoring of oral wound healing and health	CeT	LWIR		0.3	Controlled	Controlled	15
Becerra et al. [22]	2021	39	Performing repetitive tasks	ET	LWIR	0.98		20–23	<50	20
Belizer et al. [182]	2019	80	Body warmth in adolescent anorexia nervosa patients	FT, ET, NT, WT	LWIR			Controlled		15
Bhattacharjee et al. [184]	2012	17	Database	WT	LWIR					
Bhushan et al. [201]	2020	31	Emotions (guilt, shame, remorse)	FT, ET, NT, CeT, MT	MWIR		0.61	26		20
Briggs et al. [137]	2017	3	Facial cooling in outdoor windy winter conditions	NT	LWIR			−14–1		
Brioschi et al. [24]	2023	2558	Method for COVID-19 contagious people screening	FT, ET, WT	LWIR	0.98	0.3	22–23	60	15

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Table A.10 (continued).

Authors	Year	N	Topic	ROIs	Imager type	ϵ	d_{meas} [m]	AT [°C]	RH [%]	t_{acc} [min]
Buddharaju et al. [286]	2007	138	Face recognition	WT	MWIR					
Budzan et al. [191]	2013		Localisation algorithm for measuring inner canthus temperature	ET, WT	LWIR	0.98	0.5–3	23.3	44.0	≥ 15
Burzo et al. [202]	2019	14	Detecting affective state while being exposed to visual narratives	WT	LWIR					
Canavan et al. [310]	1995	44	Assessment of temporomandibular joint dysfunction	CeT	MWIR			21–23		15
Cannas Aghedu et al. [203]	2020	44	Emotions (love)	NT, MT	LWIR		1	22	50–60	15
Cardone et al. [204]	2020	10	Driver stress state evaluation	FT, NT	LWIR		1	23	50–60	20
Cardone et al. [66]	2021	6	Warping procedure for facial thermograms	WT	LWIR	0.98	1			
Cardone et al. [76]	2022	26	Drivers' mental workload estimation	FT, NT	LWIR		0.6	23	50–60	15
Cardone et al. [77]	2023	63	Facial physiology during resting state	FT, NT, CeT, GiT	LWIR			23	50–60	15
Caroccia et al. [25]	2021	46	Pain in the sternocleidomastoid muscle	FT, CeT, GiT	LWIR		1	23 ± 0.5	40–50	20
Cennamo et al. [323]	1988	77	Thermography in classic and common migraine	FT, ET, NT, CeT	MWIR					15
Cennamo et al. [324]	1990	95	Thermography in monolateral exophthalmos	ET	MWIR					15
Chan et al. [277]	2004	176	Fever screening	FT, NT, CeT, MT, WT, ErT	LWIR	0.98	1.5	27 (USC); 21 (clinic)		
Chan et al. [134]	2013	1517	Fever screening	FT, NT, CeT, MT, WT, ErT	LWIR		1.5; 1–5	20–27	Recorded	
Chan et al. [26]	2022	176	Comparison of thermal imaging and standard thermometer	WT	LWIR		2.2	22		
Chauvin et al. [173]	2016	15	Respiration rate monitoring	NT, MT	LWIR		1; 1.5; 1.75; 2; 2.5	22		
Che et al. [78]	2021	30	Effect of social touch on pain-evoked facial temperature changes	FT, ET, NT, CeT, MT	LWIR		1	26		
Chen et al. [190]	2015		Thermogram sequence registration analysis and verification	ET, WT	LWIR		1.5	26–28		
Chen et al. [174]	2020	36	Breathing rate measurement under realistic conditions	NT, MT	LWIR			Controlled		
Chen et al. [67]	2021	9	Heart rate estimation for hospitalised newborns	WT	LWIR		0.25–0.36			
Chiu et al. [278]	2005	72 327	Fever screening	WT	LWIR					
Cho et al. [205]	2019	17	Detection of perceived mental stress	NT	LWIR	0.98				
Choi et al. [206]	2015	16	Emotions (evaluation of fear)	FT, ET, CeT	LWIR			26.6	51.9	
Choi et al. [64]	2023	37	Respiration rate measurement	NT	LWIR		0.5–0.7	20–24	50–55	
Christensen et al. [175]	2012	62	Gender differences in skin temperature	CeT	LWIR	0.98	0.25	Recorded		10
Christensen et al. [138]	2014	261	Assessment of postoperative inflammation	CeT	LWIR	0.98	0.25	Not controlled		≥ 10
Chudecka et al. [27]	2022	1	Monitoring convalescence after facelift surgery	FT, ET, NT, CeT, GiT	LWIR	0.98	1.5	25 ± 0.5	55–60	20

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Table A.10 (continued).

Authors	Year	N	Topic	ROIs	Imager type	ϵ	d_{meas} [m]	AT [°C]	RH [%]	t_{acc} [min]
Clark et al. [332]	1980	17	Neonatal skin temperature distribution	FT, WT	MWIR			32.25–32.8 (incubator)		
Clark et al. [276]	2007	24	Food allergy diagnostics	FT, NT, MT	LWIR		1.5	23		
Clemente et al. [139]	2020	77	Assessment of muscular hyperactivity associated to musical performance	FT, CeT	LWIR	0.98	1	21 ± 1	47 ± 2	15
Cohen et al. [120]	2013	34	Early detection of mucositis	ET, CeT, CiT	LWIR		1	23–24		5
Cosma et al. [207]	2019	33	Thermal comfort assessment	FT, ET, NT, CeT, MT, CiT, WT	LWIR			20.0–28.8	45–55	30
Costa et al. [112]	2013	64	Reliability of IR image analysis in temporomandibular disorder	FT, CeT	LWIR	0.98	1	22		20
Costa et al. [28]	2023	11	Core temperature measurement in sleep-deprived individuals	ET	LWIR	0.98	0.7	24 ± 0.5	60–70	≥10
Cruz-Albarran et al. [208]	2017	44	Emotions detection	FT, NT, CeT, MT	LWIR	0.98	1.2	20 ± 2	45–60	15
Cruz-Albarran et al. [209]	2020	30	Acute social stress impact on facial temperature	FT, NT, CeT, CiT	LWIR		1.2	20 ± 2	45–60	15
Curzio et al. [79]	2022	12	Psychological and physiological stress in adolescents with Anorexia Nervosa	FT, NT	LWIR					
Ćosić et al. [69]	2022	30	Facial thermal map of healthy people	FT, ET, CeT, MT, CiT	LWIR		1	23–24	50–70	
de Brito Bitencourt et al. [23]	2021	32	Assessment of orofacial pain	CeT	LWIR		0.5	20–22	50–70	
De Meneck et al. [29]	2023	36	Measuring body temperature in afebrile children	ET	LWIR	0.95	0.3	23 ± 1	60	15
De Stefani et al. [210]	2019	13	Processing of facial expressions of emotions	FT, NT, CeT	LWIR		1	23 ± 1	50–55	10–20
Di Credico et al. [68]	2022	32	Estimation of heart rate variability	FT, NT	LWIR		0.6			15
Di Giacinto et al. [211]	2014	20	Emotions (fear conditioning)	NT	LWIR			23	50–60	10–20
Di Stasi et al. [80]	2022	26	Influence of emotional arousal on military performance	NT	LWIR		2.2	Monitored		20
Diaz-Piedra et al. [212]	2019	12	Emotions (arousal level)	FT, NT	LWIR		1.4	Monitored		15
Dibai Filho et al. [113]	2013	129	Diagnosis of Myogenous temporomandibular disorder	FT, CeT	LWIR	0.98	1	22		20
DiLeo et al. [140]	2017	18	Effect of wearing an N95 respirator on brain temperature measurements	FT, ET	MWIR	0.98	0.5	24.2 ± 2.6	20.5 ± 5.5	10
Drummond et al. [328]	1983	107	Source of pain in migraine headache	FT	MWIR			22 ± 1		20
Drummond et al. [322]	1984	186	Facial temperature in migraine, tension-vascular and tension headache	FT, ET, CeT	MWIR			21 ± 1		>20
Drummond [321]	1985	18	Asymmetry in chronic paroxysmal hemicrania	FT, ET, NT, CeT	MWIR			21 ± 1		>20
Drummond et al. [329]	1988	9	Atypical facial pain	FT, ET, CeT	MWIR			22 ± 1		≥20
Drummond et al. [327]	1990	80	Assessment unilateral migraine headache	FT, ET, CeT	MWIR			22 ± 1		20

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Table A.10 (continued).

Authors	Year	N	Topic	ROIs	Imager type	ϵ	d_{meas} [m]	AT [°C]	RH [%]	t_{acc} [min]
Dwijayanti et al. [30]	2023		Facial recognition and body temperature measurements	WT	LWIR		0.35–0.40			
Ebisch et al. [213]	2012	12 dyads	Mother and child synchrony	NT, MT	LWIR			23	50–60	10–20
Endo et al. [141]	2019	20	Facial temperature in patients undergoing orthognathic surgery	CeT	LWIR	0.98	0.5	22		15
Engert et al. [214]	2014	15	Stress	FT, ET, NT, MT, CiT	LWIR					5
Esposito et al. [215]	2014	33; 32	Physiological responses to infants and cognitive judgments	NT	LWIR		1	25	55	
Esposito et al. [216]	2015	38	Emotional responses to infants	NT	LWIR		1	25	55	
Fei et al. [297]	2010	20	Breathing measurement	NT	MWIR					
Fernandes et al. [116]	2016	12	Validity of inner canthus temperature	ET	LWIR	0.98				
Fernández et al. [81]	2023	56; 36	Social anxiety	FT, NT, CeT	LWIR		0.6–1	22	50	10
Ferreira et al. [31]	2021	31	Obstructive sleep apnoea	FT, ET, MT, ErT	LWIR	0.98	0.7; 1	22		15
Fikáčková et al. [289]	2006	1	Treatment of temporomandibular joint	CeT	LWIR			20		
Filippini et al. [217]	2019	17	Affective computing for educational robot	FT, NT, MT	LWIR					
Filippini et al. [82]	2021	31	Human-social robot interaction (classifying emotional engagement of the child)	NT	LWIR			23 ± 1	50–60	15
Filippini et al. [83]	2022	16	Autonomic response of newborn babies during interaction with robot	FT, NT	LWIR			23 ± 1	50–55	
Filippini et al. [84]	2022	50	Emotion recognition	FT, NT, MT	LWIR	0.98	0.6	23 ± 1	50–55	10
Ford et al. [312]	1997	1023	Diagnosis of headache	WT	LWIR			20	Controlled	20
Fournet et al. [181]	2013	18	Body temperature distribution during running	WT	LWIR	0.98	1.9	9.9 ± 0.5	54 ± 6	10
Fracasso et al. [32]	2023	24	Facial thermography pattern in migraine	FT, ET, NT, CeT, MT	LWIR	0.98	1	23 ± 1		
Fricova et al. [142]	2018	1	Monitoring of orofacial pain syndrome	WT	LWIR			23.4–23.7	59.9–62.4	20
Fricova et al. [33]	2021	30	Orofacial pain monitoring	CeT	LWIR			23.4–23.7		20
Gálvez-García et al. [85]	2021	30	Deception detection	FT, NT, CeT	LWIR		0.6–1	22	50	10
Gane et al. [218]	2011	11	Emotions (startle)	ET	LWIR		1	Recorded	Recorded	15
Ganesh et al. [86]	2021	100	Autism spectrum disorder detection	FT, ET, NT, CeT	LWIR	0.98	1.5	20	40–45	15–20
Gao et al. [34]	2022	184	Assessment of metabolic syndrome	WT	LWIR	0.98	2	22.0 ± 2.0	60–70	15
Garbey et al. [299]	2007	34	Cardiac pulse measurement	FT	MWIR		1.8			
Gasparro et al. [87]	2021	60	Detection of dental anxiety in oral surgery	FT, NT, CeT, CiT, ErT	LWIR		0.3	18–25	Controlled	
Gioia et al. [88]	2023	30	Stress	FT, NT, CeT	LWIR	0.98	1	24–26	40–60	10

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Table A.10 (continued).

Authors	Year	N	Topic	ROIs	Imager type	ϵ	d_{meas} [m]	AT [°C]	RH [%]	t_{acc} [min]
Goldman et al. [170]	2012	17	Nasal airflow and thoracoabdominal motion	NT	LWIR		0.7–1.2	22–24		
Gonzalez Viejo et al. [219]	2019	30	Sensory analysis (consumer acceptance test)	ET	LWIR			24–25		
Goulart et al. [220]	2019	28	Emotions (disgust, fear, happiness, sadness, surprise)	FT, ET, NT, CeT, MT, CiT	LWIR		0.85	22		20
Goulart et al. [221]	2019	17	Emotions recognition in a child-robot interaction (CRI)	FT, ET, NT, CeT, MT, CiT	LWIR		0.7	20–24		10
Gratt et al. [313]	1996	328	Orofacial pain assessment	WT	MWIR			20–22		15
Guan et al. [121]	2012	60	Bell's palsy treatment by acupuncture	FT, ET, NT, CeT, MT	LWIR			24–26	60	15
Gunaratne et al. [222]	2019	63	Effects of imagery on the physiological and emotional responses	ET	LWIR		0.3–0.4	24–25		
Gunaratne et al. [223]	2019	45	Physiological responses to basic tastes	ET	LWIR			24–25		
Guzman et al. [143]	2012	13	Facial signature authentication	WT	MWIR		1	23		
Haddad et al. [183]	2016	161	Heat distribution on facial skin surface	FT, ET, MT, ErT	LWIR		0.71	22	<60	15
Hahn et al. [224]	2012	16; 23	Thermal signature in social contact	FT, ET, NT, CeT, MT	LWIR	0.98	1; 0.5			20
Hastings et al. [130]	2020	5	Thermal stress in outdoor work environments	FT, NT, CeT, CiT, ErT	LWIR		2–3	30.1–39.8	24.8–58.9	
Hauvik et al. [179]	2012	12	Thermographic mapping of the skin surface of the bald head	WT	LWIR	0.98	1.5	22–23	10–20	≥15
Hong et al. [225]	2016	41	Real-time stress assessment	FT, MT	MWIR					
Hong et al. [226]	2020	60	Physical stress assessment	FT, NT, CeT, MT	MWIR					
Howell et al. [287]	2009	41	Assessment of skin temperature and blood flow in childhood localised scleroderma	FT, CeT	LWIR					15
Hu et al. [162]	2017	11	Breathing measurement	NT, MT	LWIR		1.5	not controlled		
Hu et al. [161]	2018	12	Breathing and heart rate monitoring	NT	LWIR		1–3			10
Huggins et al. [117]	2019	16	Pain and stress measurement	ET	LWIR	0.98	1			
Intharachathorn et al. [89]	2023	32	Thermal comfort assessment	FT, NT, CeT, CiT, WT	LWIR	0.98	2	21–29	51.6 ± 10.0	
Ioannou et al. [227]	2013	15	Emotions (guilt in children)	NT	LWIR			23	50–60	10–20
Ioannou et al. [228]	2014	18	Direct gaze and interpersonal proximity	FT, ET, NT, CeT, MT, CiT, WT	LWIR		0.5	20–21	60–65	15
Ioannou et al. [229]	2016	13	Sympathy crying	FT, ET, NT, CeT, MT, CiT	LWIR		1	20–21	60–65	5
Ioannou et al. [119]	2017	22	Blushing	FT, ET, NT, CeT, MT, CiT	LWIR		0.5			10–15

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Table A.10 (continued).

Authors	Year	N	Topic	ROIs	Imager type	ϵ	d_{meas} [m]	AT [°C]	RH [%]	t_{acc} [min]
Ioannou et al. [90]	2021	10	Infants thermal reactions in social interactions	FT, ET, NT, CeT, MT, CiT	LWIR	0.98	0.5	23.4	40	10
Irim et al. [135]	2020	20	Evaluation of facial tissue characteristics	FT, CeT	LWIR	0.98	1	Controlled		15
Ishimaru et al. [144]	2020	30	Diagnosis of acute inflammation in the paranasal sinus	NT, CeT	LWIR					
Iwashita et al. [71]	2021	1	Measurement of resting blood pressure	WT	LWIR	0.98	1	21.0 ± 0.6		15
Jagadev et al. [164]	2020	50	Respiration monitoring	NT	LWIR		1	Controlled		
Jakkaew et al. [165]	2020	16	Respiration monitoring	WT	LWIR	0.97	1			
Jaramillo-Quintanar et al. [70]	2022	196	Facial thermal map of Mexican population in the basal state	FT, ET, NT, CeT, CiT	LWIR	0.98	1.2	20 ± 2	45–60	10
Jaramillo-Quintanar et al. [91]	2023	25	Stress detection	FT, NT, CeT, CiT	LWIR	0.98	1.2	20 ± 2	45–60	10
Jarlier et al. [187]	2011	4	Thermal analysis of facial muscles contractions	WT	LWIR	0.98		20–23		15
Jędzierska et al. [36]	2021	10	Tonometry (intraocular pressure measurement)	FT, CiT	LWIR	0.97	0.5	22	Recorded	15
Jenkins et al. [301]	2009	16	Emotions (affective experience)	FT	MWIR	0.98	1.5			10
Jeong et al. [92]	2022	20	Emotional arousal (disabled patients)	NT	LWIR					
Jian et al. [230]	2017	35	Facial expression of schizophrenic patients	FT, NT, CeT, MT	LWIR			26–28		
Jiang et al. [35]	2021	26	Functional nasal adequacy determination	NT	LWIR		0.05–0.2	20–22		
Jukiewicz et al. [93]	2021	14	Users' emotional reaction to visual stimuli	ET, WT	LWIR	0.98	1	21 ± 1		10
Kaklauskas et al. [231]	2019	15 875; 10	Crowd emotions and decision making	FT, NT, CeT, WT	LWIR	0.98	≥5; 0.5–0.7	20		
Kargel [293]	2005	6	Measurement of the local temperature rises caused by mobile phones	CeT, ErT	LWIR		2.5	Controlled		15
Kasprzyk-Kucewicz et al. [37]	2021	27	Localisation of inflammation after surgical removal of molar teeth	CeT	LWIR		0.9 ± 0.1	19 ± 1	43 ± 5	20 ± 5
Khan et al. [294]	2006	21	Classifying facial expressions	WT	LWIR					
Khan et al. [300]	2009	23	Classifying pretended and evoked facial expressions	WT	LWIR			19–22	Controlled	20
Khan et al. [232]	2017	19	Assessment of affect and arousal level	WT	LWIR			19–22		≥20
Kim et al. [133]	2019	508	Correlation between tympanic membrane and facial temperature	FT, ET	LWIR		1.5	26.0 ± 1.5; 26.2 ± 1.3	42.3 ± 16.8; 43.8 ± 17.4	
Kim et al. [146]	2019	32	Assessment of transnasal sphenopalatine ganglion block	FT, CeT, CiT	LWIR			24		30
Knapik et al. [233]	2019	3	Driver's fatigue recognition	ET, MT, WT	LWIR			22; 10		
Kobel et al. [296]	2003	10	Face recognition based on facial thermal pattern	WT	LWIR			21.5	28	15
Kolosovas-Machuca et al. [177]	2011	25	Skin temperature distribution in Mexican children	FT, CeT, CiT	LWIR	0.97	1.5	22 ± 1	40	15

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Table A.10 (continued).

Authors	Year	N	Topic	ROIs	Imager type	ϵ	d_{meas} [m]	AT [°C]	RH [%]	t_{acc} [min]
Kopaczka et al. [192]	2018	90	Facial thermal imaging database	WT	LWIR		0.9			
Koroteeva et al. [94]	2022	25	Emotional sweating	FT, NT, CeT	MWIR	0.98	1	20–22	65–70	10–15
Kosonogov et al. [234]	2017	24	Emotional arousal	NT	MWIR	0.98		24		20
Koukiou et al. [136]	2015	41	Drunk person identification	WF	LWIR		0.3	Controlled		
Kretschmer et al. [95]	2023	12	Facial thermal response to non-painful stressor (hunger) in premature and term neonates	FT, ET, NT, MT, CiT	LWIR		0.4	23–25		30
Krzywicki et al. [128]	2014	22	Measuring eccrine sweat gland activity	WF	MWIR		2.3	25.5–27.7	33.5–43.7	10
Kubicek et al. [129]	2019	20	Prediction model of alcohol intoxication	FT, NT	LWIR	0.96	2	22	55	30
Kuderer et al. [180]	2020	41	Warmth in adolescents with Anorexia Nervosa	FT, ET, NT, CeT, MT, CiT	LWIR		2	18–25		10
Kudrow et al. [330]	1985	462	Thermal pattern in headache	FT	MWIR		0.6	21.1		10
Kwon et al. [57]	2021	107	Respiratory rate monitoring	NT	LWIR		1			
Kwon et al. [63]	2023	15	Breathing monitoring	NT	LWIR		0.5			
Lahiri et al. [188]	2015	5	Mobile phone induced heating	CeT, ErT	MWIR	0.98	0.4	28 ± 1	Controlled	20
Lai et al. [96]	2021	25	Attention-Deficit Hyperactivity Disorder emotion recognition	WT	LWIR					
Larbig et al. [290]	2006	30	Assessment of antihistaminic activity	NT	LWIR		1.2	22 ± 0.4	50 ± 6	30
Latif et al. [235]	2016		Emotion detection	FT, ET, NT, MT	LWIR	0.98	1	24	50	
Legrand et al. [178]	2015	92	Facial temperature during exercise	FT, CeT	MWIR		2	19–20		
Levine et al. [302]	2009	11	Workplace stress	FT	MWIR					
Lewis et al. [168]	2011	25	Measurement of respiration rate and relative tidal volume	NT	LWIR, MWIR			21–22	Controlled	
Li et al. [236]	2018	12	Thermal comfort assessment	FT, NT, CeT, MT, ErT	LWIR		1	22.5–27.7	20.2–33.4	10
Li et al. [97]	2021	41	Method to detect clothing level and facial skin temperature for controlling HVAC system	WT	LWIR		0.3	22.6–24.3	Measured	10
Li et al. [98]	2023	24	Analysis of emotional experiences ob B2C websites	FT, NT, CeT, MT	LWIR		1			15
Liu et al. [148]	2020	26	Analysis of facial acupoints of patients with facial paralysis	FT, CeT, MT	LWIR	0.98	1.2	23–24	50–60	20
Liu et al. [38]	2021	90	Facial paralysis detection	FT, ET, NT, CeT, MT	LWIR	0.98	1.2	24–25	45–55	15
Liu et al. [39]	2021	30	Bell's palsy assessment	FT, ET, NT, CeT, MT	LWIR	0.98	1.2	23–25	45–60	30
Lorato et al. [58]	2021	15	Monitoring respiratory flow and motion	NT	LWIR					

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Table A.10 (continued).

Authors	Year	N	Topic	ROIs	Imager type	ϵ	d_{meas} [m]	AT [°C]	RH [%]	t_{acc} [min]
Lyra et al. [59]	2021	26	Monitoring the respiratory rate of patients in the intensive care unit	WT	LWIR			20–27.5		
Mačianskytė et al. [40]	2022	46	Automatic detection of maxillofacial tumors	WT	LWIR	0.98	1	22 ± 1	60 ± 5	15
Makino Antunes et al. [41]	2023	136	Detection of SARs diseases	ET, NT, MT	LWIR		0.8	22.2 ± 0.8	60.5 ± 2.6	
Manini et al. [237]	2013	14	Mother and child (synchrony)	NT	LWIR			23	50–60	10–20
Marinescu et al. [238]	2016	14	Mental workload estimation	ET, NT, CeT	MWIR			Recorded	Recorded	
Marinescu et al. [239]	2018	14	Mental workload estimation	FT, ET, NT, MT	MWIR					
Marqués-Sánchez et al. [240]	2020	1	Emotional competences	FT, NT, CeT	LWIR	0.96	0.8	18.7–24.4	26.1–42.2	10–15
Marzec et al. [149]	2015	98	Detection of characteristic facial areas in thermogram	FT, ET, NT, MT	LWIR		1	21	45–55	
Masaki et al. [42]	2021	1	Anomaly detection in facial skin temperature	WT	LWIR	0.98	1	24.7 ± 6.0		
Masaki et al. [99]	2022	1	Detection of drowsiness in drivers	WT	LWIR	0.98	1	Monitored		>30
Mauriz et al. [241]	2020	40	Perception of anxiety	FT, ET, NT, MT	LWIR		1	Monitored	Monitored	
Mazzone et al. [242]	2017	28	Physiological response to social exclusion in early adolescence	NT	LWIR			23 ± 1	50–55	10
McBeth et al. [315]	1996	59	Assessment of temporomandibular disorders	CeT, ErT	MWIR			20–22		15
Memarian et al. [285]	2009	10	Access pathway (local temperature changes associated with mouth opening/closing to enable binary switch)	MT	LWIR	0.98				
Metzmacher et al. [243]	2018	1	Thermal comfort	FT, ET, NT, CeT, MT, CiT	LWIR		1.4	17–25	32	
Micheletti et al. [43]	2022	20	Relative stability of temperature measurements	FT, ET	LWIR		0.15	21.1 ± 1.7	60.9 ± 1.9	
Mijović et al. [127]	2012	9	Sweat evaporation inside clothing systems	FT	LWIR					
Mizukami et al. [333]	1987	21	Emotional behavioural research in infants	FT	LWIR		1–1.5	25		
Mizuno et al. [244]	2012		Colour influences on human	FT, NT	LWIR	0.98	1			
Mizuno et al. [245]	2016	9	Mental workload estimation	FT, NT, CeT, MT	LWIR		1	Controlled	Controlled	
Mo et al. [62]	2022	15	Respiration rate monitoring	NT	LWIR			23–26		
Moliné et al. [246]	2017	31; 20; 20	Deception detection, cold stress test	FT, ET, NT, CeT, MT	LWIR		0.6	22	50	10
Moliné et al. [247]	2018	20; 20; 20; 20	Thermal markers of phobia and anxiety, lying, mental effort, real vs. simulated emotions	FT, ET, NT, CeT, MT	LWIR		0.6–1	22	50	10
Mongini et al. [316]	1993	20	Efficacy of meclofenamate sodium versus placebo in headache and craniofacial pain	FT, CeT	LWIR			21 ± 1		15

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Table A.10 (continued).

Authors	Year	N	Topic	ROIs	Imager type	ϵ	d_{meas} [m]	AT [°C]	RH [%]	t_{acc} [min]
Moreira et al. [44]	2021	47	Assessment of the thermal effect of prosthodontic treatment on the cranio-cervico-mandibular complex	FT, CeT, MT	LWIR	0.98	1			15
Morresi et al. [100]	2023	15	Thermal comfort assessment	FT	LWIR	0.99	1.5	22–28	35–55	
Muniz et al. [45]	2022	198	Screening of febrile people	FT, ET, ErT	LWIR	0.98	1.5			
Murty et al. [298]	2009	27	Airflow monitoring during polysomnography	NT	MWIR					
Mutlu et al. [172]	2018	6	Respiration rate monitoring	NT	LWIR		0.6			
Nakamura et al. [282]	2006	10	Appropriate construction bite elucidation	FT, CeT, MT, CiT	MWIR			25 ± 1	53 ± 1	>30
Nakane et al. [159]	2020	2	Blood pressure measurement	WT	LWIR	0.98	0.7	25.0 ± 1.0		15
Nakanishi et al. [303]	2008	12	Emotions (joyful expression in infants)	FT, NT, CeT	MWIR		2	25	55	
Negishi et al. [167]	2020	63	Respiration rate (vitals signs measurement system)	NT, MT, WF	LWIR		1			
Neves et al. [176]	2017	94	Body temperature distribution	FT, ET, NT, CiT, WT	LWIR	0.98		23 ± 1.0		15
Ng et al. [279]	2004	502	Fever screening	FT, ET	LWIR	0.98	2			
Ng et al. [291]	2006	502	Fever screening	FT, ET	LWIR	0.98	1.5			
Nguyen et al. [280]	2010	3000	Fever screening	WT	LWIR		2–3	Recorded		
Nhan et al. [304]	2009	12	Emotions (classifying affective states)	FT, ET, NT	LWIR	0.98	1	Controlled	Controlled	≥ 20
Nhan et al. [283]	2009	12	Baseline characteristics of facial skin temperature	FT, ET, NT	LWIR	0.98	1	24.9 ± 1.7	32.1 ± 8.1	20
Nicolini et al. [248]	2019	24	Autonomic responses to emotional stimuli in children affected by facial palsy	NT	LWIR		1	23	50–60	10
Niedzielska et al. [151]	2017	39	Assessment of paranasal sinus pathology	CeT	LWIR		0.4–1	22–23	50–55	15
Novikov et al. [60]	2021	1	Assessment of nasal breathing efficiency in infants and young children with unilateral cleft lip	NT	LWIR		0.2–0.4			
Nozawa et al. [305]	2009	4	Arousal level	NT	LWIR	0.98	1	23 ± 1		
Oiwa et al. [160]	2018	8	Blood pressure measurement	FT, NT, CeT	LWIR		0.6	25 ± 1		
Okamoto et al. [152]	2016	4	Thermal physiology of the facial skin	WT	LWIR	0.98	0.8	25 ± 1.0		≥ 15
Okur Güney et al. [249]	2015	2 couples	Embodied interpersonal emotion regulation in somatic symptom disorders	FT, NT	LWIR					
Or et al. [306]	2007	33; 13	Cognitive load	FT, NT	LWIR			22; 23		5–10
Osia et al. [185]	2014	138	Face recognition	WT	LWIR, MWIR		1.5			
Padilla-Medina et al. [111]	2014	8	Acne bulgaris assessment	CeT	LWIR	0.98	0.6	25 ± 1		30
Panasiti et al. [250]	2016	12	Deception detection	ET, NT, CeT	LWIR		1	23	50–60	20
Panasiti et al. [251]	2019	33	Cognitive load and emotional processing in psoriasis	ET, NT	LWIR		1	23 ± 1		30
Panasiuk et al. [252]	2020		Study on the influence of psychophysical stimuli on facial skin temperature	FT, ET, NT, CeT	LWIR		1.5	22	35	

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Table A.10 (continued).

Authors	Year	N	Topic	ROIs	Imager type	ϵ	d_{meas} [m]	AT [°C]	RH [%]	t_{acc} [min]
Panetta et al. [193]	2018	113	Multimodal dataset	WT	LWIR		Controlled			
Paoletti et al. [101]	2022	21	Abstract words processing	NT	LWIR		1	20–22	40	15
Paolini et al. [253]	2016	24; 21	Ostracism	NT, MT	LWIR			23 ± 1	50–55	10–15
Park et al. [254]	2013	30	Deception detection	ET	LWIR		1	21 ± 0.25	65 ± 2	2
Pavlidis et al. [307]	2002	32	Lie detection	FT, ET	MWIR					
Pavlidis et al. [308]	2007	12	Stress (HCI system)	FT	MWIR			19.4–22.2		
Pavlidis et al. [255]	2012	17	Stress (fast behaviour)	NT	MWIR		2.4			
Pavlidis et al. [256]	2016	78	Driver behaviours under stressors	MT	LWIR					
Pavlidis et al. [153]	2019	18	Quantification of migrainous thermal facial patterns	FT, ET	MWIR					
Peko Cohen et al. [186]	2019	2	Assessment of facial tissue stress	FT, CeT, GiT	LWIR			24–25		
Perpetuini et al. [72]	2021	16	Improving fNIRS method	WT	LWIR		0.6	23	50–60	20
Perpetuini et al. [102]	2022	8	Brain-derived fatigue during unilateral resistance exercise	FT, NT	LWIR		0.6	22–24	50 ± 5	5
Pogrel et al. [326]	1989		Oral and maxillofacial thermography	CeT	MWIR					15
Pollina et al. [309]	2006	39; 30	Deception detection (concealed information test)	CeT, WT	MWIR		1.5	20–22		
Ponsi et al. [257]	2017	33	Social categorisation bias	ET	LWIR		1	23 ± 1		30
Ponsi et al. [258]	2019	34	Physiological and behavioural reactivity to social exclusion	ET	LWIR		1	23 ± 1		30
Rajoub et al. [259]	2014	25	Deception detection	ET	MWIR					
Rimm-Kaufman et al. [319]	1996	32	Emotions	FT, ET, CeT	MWIR		0.8	Recorded		10
Robinson et al. [260]	2012	114	Dimensional aspects of emotion (interaction)	FT, ET, CeT, MT	LWIR	0.98	1	Observed	Observed	
Rodrigues-Bigaton et al. [126]	2013	30	Diagnosis of arthralgia	ErT	LWIR	0.98	1	22 ± 1		
Rodríguez Medina et al. [261]	2018	33	Assessment of pain in breast cancer survivors	FT, ET, NT, MT, GiT	LWIR		1.5	22 ± 2		15–20
Rudokaite et al. [103]	2022	45	Vasovagal reactions in people with needle fear	FT, NT, CeT, MT, GiT	LWIR		1	21.84 ± 1.55	42.54 ± 3.15	
Rumiński [171]	2016	28	Respiration rate monitoring	NT	LWIR		0.4–0.7	22; 27		
Rusli et al. [262]	2020	57	Affective states recognition of children with Autism Spectrum Disorder	ET	LWIR	0.98	1.5	24		
Rusztowicz et al. [46]	2022	14	Evaluation of the efficacy of cosmetic acids	WT	LWIR			20–21	40–50	30
Rytivaara et al. [47]	2021	40	Facial pain	FT, CeT	LWIR		1	22		30

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Table A.10 (continued).

Authors	Year	N	Topic	ROIs	Imager type	ϵ	d_{meas} [m]	AT [°C]	RH [%]	t_{acc} [min]
Salazar-López et al. [263]	2015	120; 50; 30	Emotions (valence and arousal, empathy, subjective experiences)	FT, ET, NT, CeT, MT	LWIR		0.6	22	50	10
Sathyamoorthy et al. [104]	2023	50	Emotion detection	FT, ET, NT, CeT	LWIR		1	20		
Sato et al. [264]	2020	20	Physiological correlates of subjective emotional valence and arousal dynamics	NT	LWIR			23.5–24.5		10
Scarano et al. [154]	2018	15	Postoperative pain assessment (lateral sinus floor elevation)	CeT	LWIR		0.5	22–24	50 ± 5	
Scarano et al. [265]	2020	20	Discomfort when wearing protective face masks	MT	LWIR	0.98	0.5	22–24	50 ± 5	
Scarano et al. [158]	2020	23	Postoperative pain assessment (eradication of benign skin lesions)	NT	LWIR		0.5	22–24	50 ± 5	
Scarano et al. [155]	2020	45	Postoperative pain assessment (treatment of nasal telangiectasia)	WT	LWIR		0.5	22–24	50 ± 5	
Scarano et al. [49]	2021	135	Post-operative pain assessment (treatment of static crow's feet)	ET	LWIR		0.5	22–24	50 ± 5	
Schöffauer et al. [48]	2023	30	Facial thermal asymmetries in dental thermography	CeT, MT, WT	LWIR	0.98	0.15; 0.3; 0.5	21.4	27	
Seidel et al. [266]	2020	103	Vascular response to social cognitive performance	NT, CeT	LWIR		0.32	22.94 ± 1.14	59.62 ± 7.34	15
Seppey et al. [317]	1993	22	Nasal provocation tests with histamine and allergen	NT	MWIR		1	20	50	30
Septiadi et al. [267]	2019	81	Anxiety assessment	FT	MWIR		0.1	24		
Sharma et al. [268]	2014	35	Stress recognition	WT	LWIR					
Shastri et al. [281]	2008	10	Neurophysiological responses	FT, ET, MT	MWIR		2.7			10
Shastri et al. [269]	2012	18; 24	Physiological stress	NT, MT	MWIR		2.4			
Singh et al. [156]	2017	100	Paranasal sinusitis detection	NT, CeT, MT	LWIR	0.98	1	22 ± 5	55 ± 10	
Singh et al. [131]	2018	70	Thermographic evaluation on orofacial regions	NT, CeT, CiT	LWIR	0.98	0.8	21 ± 3	60	
Singh et al. [50]	2021	35	Detection of maxillary sinusitis	CeT	LWIR	0.98		25 ± 0.5	60	15
Sivanandam et al. [118]	2012	62	Diabetes diagnostics	FT, ET, NT, ErT	LWIR	0.98	0.9	23	50	15
Socolinsky et al. [284]	2003	115	Face recognition	WT	LWIR			Not controlled		
Sonkusare et al. [270]	2019	20	Emotions (sudden auditory stimulus)	FT, NT, CeT	LWIR	0.98		22 ± 2	55–70	5
Sonkusare et al. [105]	2021	20	Emotion recognition	NT, WT	LWIR	0.98	1	22 ± 2	55–65	10
Speak et al. [106]	2021	775	Influence of tree shade on thermal comfort	FT	LWIR	0.98	1	20–38	20–80	
Stanić et al. [107]	2022	23	Dynamic seat assessment for enabled restlessness	FT, NT	LWIR	0.98	3	21.5 ± 1.2	38.8 ± 8.4	10–15
Stokholm et al. [51]	2021	64	Facial skin temperature in acute stroke patients with delirium	FT, ET, MT	LWIR			20.7–28.9		
Stüttgen et al. [325]	1982		Flush phenomena	WT	MWIR			Controlled		20

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Table A.10 (continued).

Authors	Year	N	Topic	ROIs	Imager type	ϵ	d_{meas} [m]	AT [°C]	RH [%]	t_{acc} [min]
Sugimoto et al. [320]	2000	5	Emotions	FT, ET, NT, CeT, MT, CiT	MWIR		1	16.9; 25.4		
Taurisano [318]	2000	2; 10	Mobile phone induced heating	FT, CeT, ErT	LWIR	0.98	0.5–2	20; 21	70; 71; 73; 77	
Telson et al. [65]	2023	20	Analysis of breathing mode	NT, MT	LWIR	0.98	1	20 ± 1	<70	15–20
Thirunavukkarasu et al. [157]	2020	200	Evaluation of type II diabetes mellitus	FT, ET, NT, CeT, CiT, WT	LWIR		1	22–23	50	15
Thukral et al. [52]	2023	50	Prediction of oral mucositis in patients with head-and-neck cancer	WT	LWIR		0.8–1.2	Recorded	Recorded	10–15
Tian et al. [73]	2023	40	Influencing factors on measurement of facial skin temperature using thermal imager	FT, NT, CeT, CiT	LWIR	0.98	1; 2; 3	12–32	60	
Torrico et al. [271]	2018	60	Physiological and affective responses of consumers	ET	LWIR					
Tyler et al. [272]	2015	19; 45; 20	Stress	FT, NT, CeT, CiT	LWIR		1.5	24		
Umapathy et al. [53]	2021	100	Orofacial pain detection	WF	LWIR		1	20	45–50	20
Vardasca et al. [189]	2017	14	Influence of angles and distance on thermal imaging	ET	LWIR		0.7–2	22.9 ± 0.3	59.0 ± 2.1	10
Volta et al. [331]	1988	32	Assessment of migraine headache	FT	MWIR					
Wagner et al. [132]	2013	1; 10	Introduction of a newly developed circular suction technique for ablative interventions	WT	LWIR					
Wang et al. [292]	2010	215	Facial expression recognition	FT, ET, NT, CeT, MT	LWIR		0.75	18–32		
Wang et al. [54]	2022	1020	Measuring elevated body temperature	FT, ET, MT, WF	LWIR	0.98	0.6–0.8	20–29	10–62	15
Wang et al. [108]	2023	32	Facial temperature distribution changes during meditation	FT, NT, CiT, WT	LWIR			23–25	50–60	15
Warmelink et al. [273]	2011	51	Lie detection	WF	LWIR		3			
Weinstein et al. [311]	1991	102; 300	Evaluation of thermographic abnormality and temporomandibular disorder	FT, ET, CeT, CiT, ErT	LWIR			20–22		15
Wilson et al. [55]	2022	66	Characterisation of baseline temperature characteristics	CeT	LWIR	0.98	0.3048	21.3–23.6	50	15
Woźniak et al. [150]	2015	100	Identification of temporomandibular disorder	FT, CeT	LWIR	0.98	1	22–24	50–70	20
Xuan et al. [56]	2021	40	Motion sickness	FT, NT, WT	LWIR	1	0.6–0.7	23 ± 1.3		15
Yoshida et al. [109]	2023	9	Drowsiness estimation	WT	LWIR	0.98	1	25.4 ± 0.8		30
Zago et al. [274]	2019	40	Detecting malingering of amnesia in crime	ET	LWIR			18–22	Recorded	
Zajonc et al. [334]	1989	26; 40; 26; 20	Emotions	FT	LWIR			20.0–21.1		10

(continued on next page)

Table A.10 (continued).

Authors	Year	N	Topic	ROIs	Imager type	ϵ	d_{meas} [m]	AT [°C]	RH [%]	t_{acc} [min]
Zaproudina et al. [295]	2008	16	Reproducibility of infrared thermography measurements	FT, CeT	MWIR	0.97	1	23–24		20
Zaproudina et al. [114]	2013	38	Asynchronicity of facial blood perfusion in migraine	NT, CeT	LWIR		2	22–24	30–50	≥ 15
Zaproudina et al. [147]	2013	11	Assessment of nitroglycerin-induced changes in facial skin temperature	NT, CeT, MT	LWIR		1	22–24	30–50	≥ 30
Zhang et al. [288]	2007	180	Selection of acupoints for treatment of peripheral facial paralysis	ET, CeT, MT	LWIR		1.5			
Zhang et al. [275]	2019	18	Discriminating drivers' emotions	FT, NT	LWIR		1–1.2			
Zhou et al. [145]	2020	596	Fever screening	FT, ET, MT, WT	LWIR	0.98	0.6–0.8	20–24	10–62	15
Zhu et al. [314]	1999	223	Distribution pattern of skin temperature in Chinese population	FT, CeT	MWIR	0.98	1.5	25	40	20



Fig. B.16. Simplified representation of (a) horizontal FOV with associated number of pixels and (b) IFOV (individual pixel size).

for the horizontal FOV and

$$d_v = 1 \text{ mm} \cdot \frac{768}{9} \cdot \frac{180}{\pi} \cdot \frac{1}{1000} = 4.9 \text{ m} \quad (\text{B.3})$$

for vertical FOV. Note that there is no difference between d_h and d_v .

Data availability

Data will be made available on request.

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